CSCI218: Foundations of Artificial Intelligence

UNIVERSITY
OF WOLLONGONG
AUSTRALIA

Outline of Probabilistic Reasoning

- Uncertainty
  - Probability
  - Bayes Nets
- Uncertainty and Time
  - Markov Models
  - Hidden Markov Models
- Summary

Uncertainty

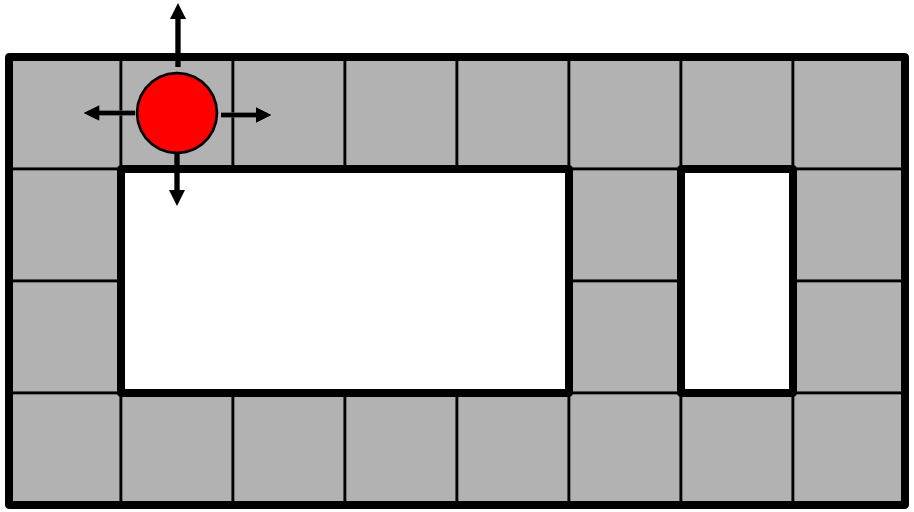
- The real world is rife with uncertainty!
  - e.g., if I leave for Sydney 2 hours before my flight, will I be there in time?
- Problems:
  - partial observability (road state, other drivers' plans, etc.)
  - noisy sensors (radio traffic reports, Google maps)
  - immense complexity of modelling and predicting traffic, security line, etc.
  - lack of knowledge of world dynamics (will tire burst? need COVID test?)
- An agent may never know for sure
  - What state it is in now
  - Where it will end up after a sequence of actions

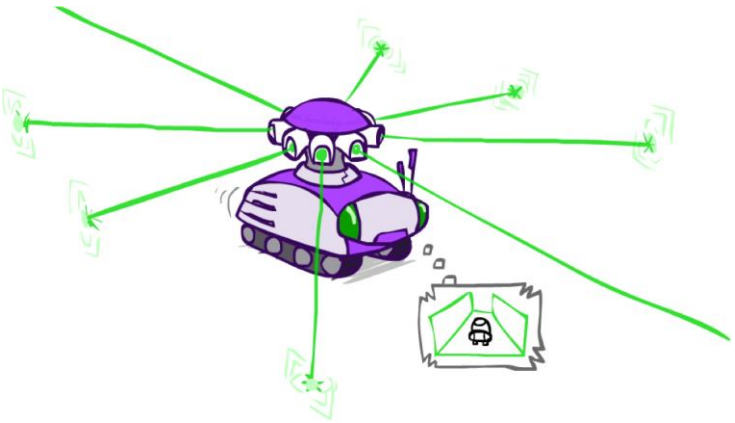
Uncertainty

- For complex tasks, we cannot afford
  - Keep track of all possible world states
  - Generate a contingency plan that handles every possibility
- **Laziness**: Too much work to list the complete set of rules
- **Ignorance**: We have no complete theory for a domain or we do not have all the needed information
- **Probability** provides a way of summarizing the uncertainty that comes from our ***ignorance*** and ***laziness***

Example: Robot Localization

Example from
Michael Pfeiffer





Prob 0 1

t=0

Sensor model: four bits for wall/no-wall in each direction, never more than 1 mistake

Transition model: action may fail with small prob.

Uncertainty

Decision theory = Probability theory + Utility theory

--- An agent is rational if and only if it chooses the action that yields the highest expected utility, averaged overall all the possible outcomes of the action.

Maximize expected utility : $a * = \ argmax_a \sum_s P(s \mid a)\, U(s)$

Uncertainty

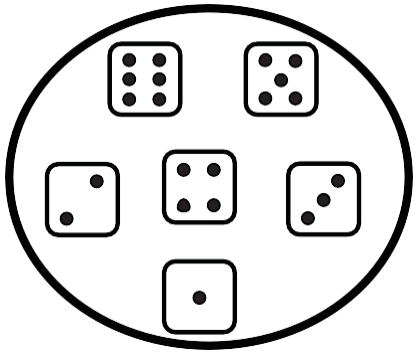
```
function DT-AGENT(percept) returns an action
  persistent: belief_state, probabilistic beliefs about the current state of the world
              action, the agent's action

  update belief_state based on action and percept
  calculate outcome probabilities for actions,
      given action descriptions and current belief_state
  select action with highest expected utility
      given probabilities of outcomes and utility information
  return action
```

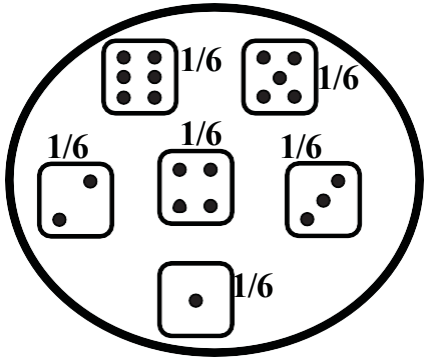
A decision-theoretic agent that selects rational action

Basic laws of probability

- Begin with a set ν of possible worlds
  - E.g., 6 possible rolls of a die, {1, 2, 3, 4, 5, 6}

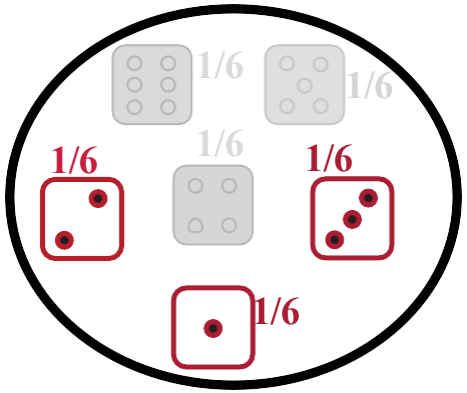


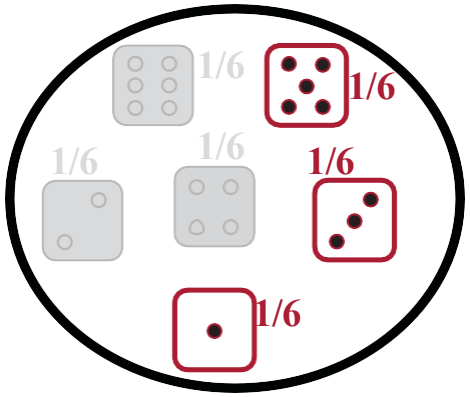
- A ***probability model*** assigns a number $P(\omega)$ to each world ω
  - E.g., $P(1) = P(2) = P(3) = P(5) = P(5) = P(6) = 1/6$.
- These numbers must satisfy
  - $0 \leq P(\omega) \leq 1$
  - $\sum_{\omega \in \Omega} P(\omega) = 1$



Basic laws contd.

- An ***event*** is any subset of Ω
  - E.g., "roll < 4" is the set {1,2,3}
  - E.g., "roll is odd" is the set {1,3,5}



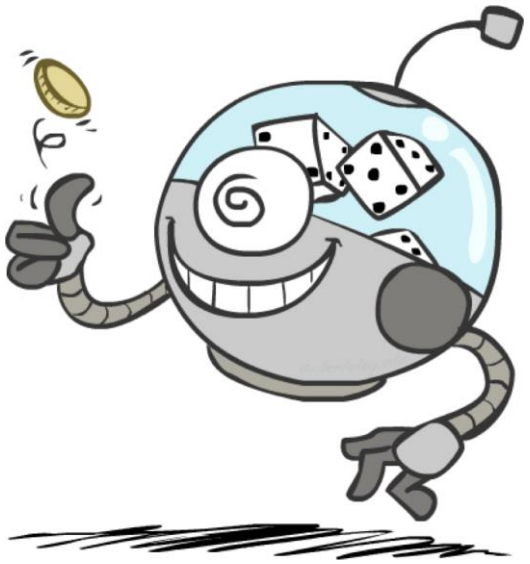


- The probability of an event A is the ***sum*** of probabilities over its worlds
  - $P(A) = \sum_{\omega \in \Omega} P(\omega)$
  - e.g., $P(roll < 4) = P(1) + P(2) + P(3) = 1/2$

- De Finetti (1931): anyone who bets according to probabilities that violate these laws can be forced to lose money on every set of bets

Random Variables

- A random variable (usually denoted by a capital letter) is some aspect of the world about which we (may) be uncertain



- The ***range*** of a random variable is the set of possible values
  - T = Is it hot or cold? ® {hot, cold}
  - D = How long will it take to get to the airport? ® $[0, \infty)$
  - L_{Ghost} = Where is the ghost? ® {(0,0), (0,1), …}

- The ***probability distribution*** of a random variable X gives the probability for each value x in its range (probability of the event $X=x$)
  - $P(X=x)$
  - $P(x)$ for short (when unambiguous)
  - $P(X)$ refers to the entire distribution (think of it as a vector or table)

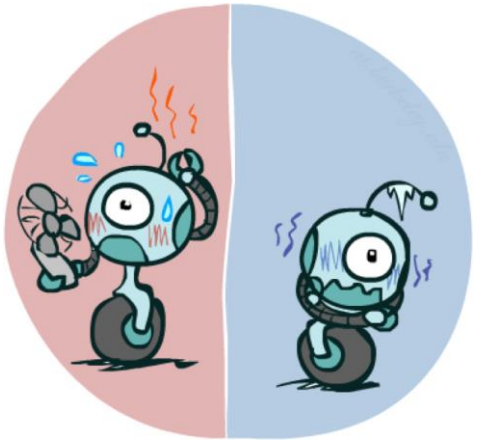
Probability Distributions

- Associate a probability with each value; sums to 1

  - Temperature:

$P(T)$

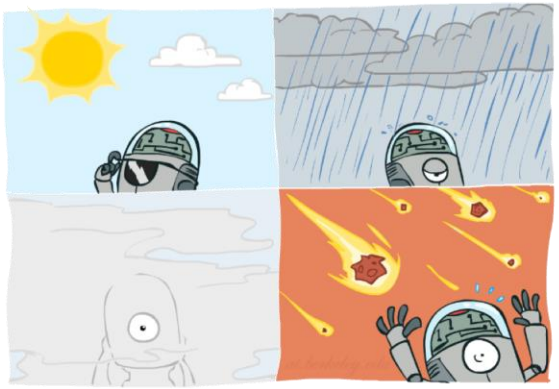
T	P
hot	0.5
cold	0.5



  - Weather:

$P(W)$

W	P
sun	0.6
rain	0.1
fog	0.3
meteor	0.0



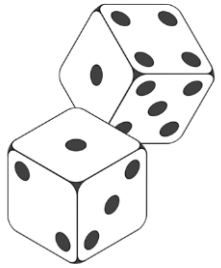
  - ***Joint distribution***

$P(T,W)$

		Temperature	
		hot	cold
Weather	sun	0.45	0.15
	rain	0.02	0.08
	fog	0.03	0.27
	meteor	0.00	0.00

Making possible worlds

- In many cases we
  - begin with random variables and their domains
  - construct possible worlds as assignments of values to all variables
- E.g., two dice rolls $Roll_1$ and $Roll_2$
  - How many possible worlds?
  - What are their probabilities?



- Size of distribution for n variables with range size d?

$$d^n$$

- For all but the smallest distributions, cannot write out by hand!

Probabilities of events

- Recall that the probability of an event is the sum of probabilities of its worlds:
  - $P(A) = \sum_{\omega \in A} P(\omega)$

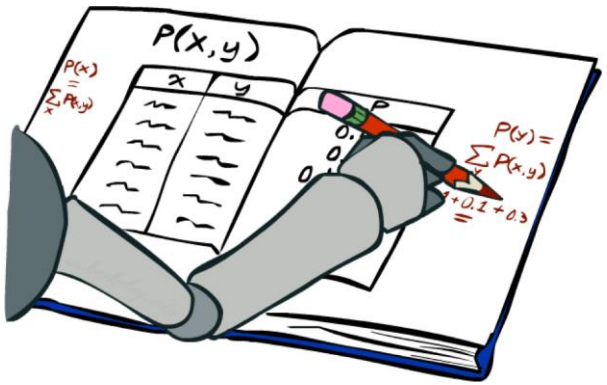
- So, given a joint distribution over all variables, can compute **any** event probability!
  - Probability that it's hot AND sunny?
  - Probability that it's hot?
  - Probability that it's hot OR not foggy?

- ***Joint distribution***

$P(T,W)$

		Temperature	
		hot	cold
Weather	sun	0.45	0.15
	rain	0.02	0.08
	fog	0.03	0.27
	meteor	0.00	0.00

Marginal Distributions



- Marginal distributions are sub-tables which eliminate variables
- ***Marginalization*** (***summing out***): Collapse a dimension by adding

$$P(X = x) = \sum_{y} P(X = x, Y = y)$$

		Temperature		
		hot	cold	
Weather	sun	0.45	0.15	0.60
	rain	0.02	0.08	0.10
	fog	0.03	0.27	0.30
	meteor	0.00	0.00	0.00
		0.50	0.50	

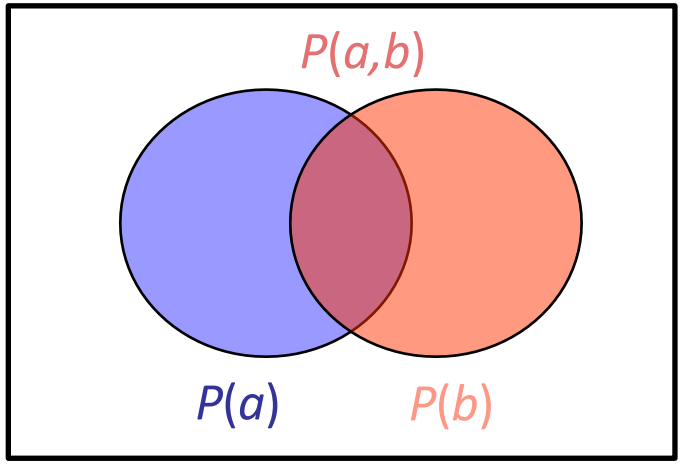
$P(W)$

$P(T)$

Conditional Probabilities

- A simple relation between **joint** and **conditional** probabilities
  - In fact, this is taken as the *definition* of a conditional probability

$$P(a \mid b) = \frac{P(a, b)}{P(b)}$$



$P(T,W)$

		Temperature	
		hot	cold
Weather	sun	0.45	0.15
	rain	0.02	0.08
	fog	0.03	0.27
	meteor	0.00	0.00

$$P(W{=}s \mid T{=}c) = \frac{P(W{=}s, T{=}c)}{P(T{=}c)} = 0.15/0.50 = 0.3$$

$= P(W{=}s,T{=}c) + P(W{=}r,T{=}c) + P(W{=}f,T{=}c) + P(W{=}m,T{=}c)$
$= 0.15 + 0.08 + 0.27 + 0.00 = 0.50$

Conditional Distributions

- Distributions for one set of variables given another set

		Temperature	
		hot	cold
Weather	sun	0.45	0.15
	rain	0.02	0.08
	fog	0.03	0.27
	meteor	0.00	0.00

$P(W \mid T=h)$

hot
0.90
0.04
0.06
0.00

$P(W \mid T=c)$

cold
0.30
0.16
0.54
0.00

$P(W \mid T)$

hot	cold
0.90	0.30
0.04	0.16
0.06	0.54
0.00	0.00

Normalizing a distribution

- (Dictionary) To bring or restore to a normal condition

All entries sum to ONE

- Procedure:
  - Multiply each entry by α = 1/(sum over all entries)

$P(W,T)$

		Temperature	
		hot	cold
Weather	sun	0.45	0.15
	rain	0.02	0.08
	fog	0.03	0.27
	meteor	0.00	0.00

$P(W,T=c)$

0.15
0.08
0.27
0.00

Normalize

$\alpha = 1/0.50 = 2$

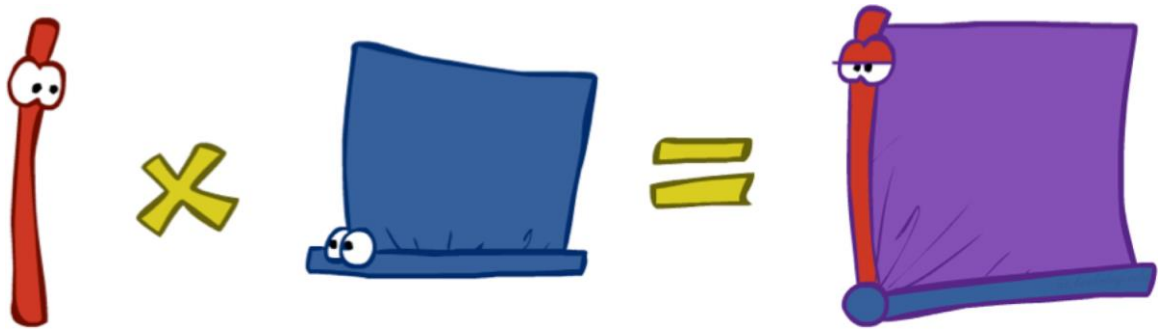
$P(W \mid T=c) = P(W,T=c)/P(T=c)$
$= \alpha\, P(W,T=c)$

0.30
0.16
0.54
0.00

The Product Rule

- Sometimes have **conditional** distributions but want the **joint**

$$P(a \mid b)\, P(b) = P(a, b) \quad \Longleftrightarrow \quad P(a \mid b) = \frac{P(a, b)}{P(b)}$$



The Product Rule: Example

$$P(W \mid T)\, P(T) = P(W, T)$$

$P(W \mid T)$

hot	cold
0.90	0.30
0.04	0.16
0.06	0.54
0.00	0.00

$P(T)$

T	P
hot	0.5
cold	0.5

$P(W, T)$

		Temperature	
		hot	cold
Weather	sun	0.45	0.15
	rain	0.02	0.08
	fog	0.03	0.27
	meteor	0.00	0.00

The Chain Rule

- A **joint** distribution can be written as a product of **conditional** distributions by **repeated** application of the product rule:
- $P(x_1, x_2, x_3) = P(x_3 \mid x_1, x_2)\, P(x_1, x_2) = P(x_3 \mid x_1, x_2)\, P(x_2 \mid x_1)\, P(x_1)$

- $P(x_1, x_2, \ldots, x_n) = \prod_i P(x_i \mid x_1, \ldots, x_{i-1})$

Probabilistic Inference

- Probabilistic inference: compute a desired probability from a probability model
  - Typically for a ***query variable*** given ***evidence***
  - E.g., P(airport on time | no accidents) = 0.90
  - These represent the agent's *beliefs* given the evidence

- Probabilities change with new evidence:
  - P(airport on time | no accidents, 5 a.m.) = 0.95
  - P(airport on time | no accidents, 5 a.m., raining) = 0.80
  - Observing new evidence causes ***beliefs to be updated***

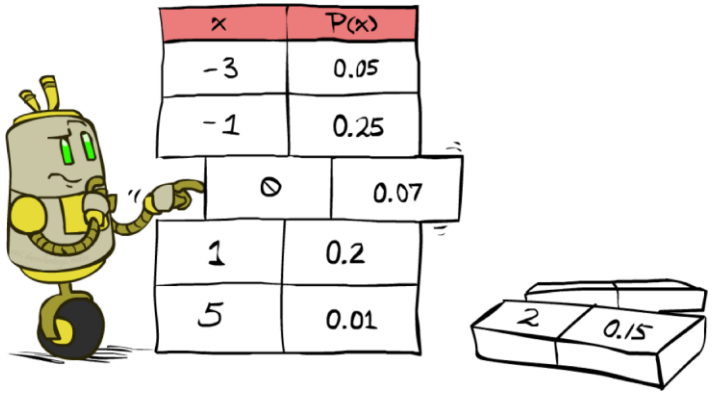
Inference by Enumeration

- Probability model $P(X_1, \ldots, X_n)$ is given
- Partition the variables $X_1, \ldots, X_n$ into sets as follows:
  - Evidence variables: $\boldsymbol{E} = \boldsymbol{e}$
  - Query variables: $\boldsymbol{Q}$
  - Hidden variables: $\boldsymbol{H}$

- We want:

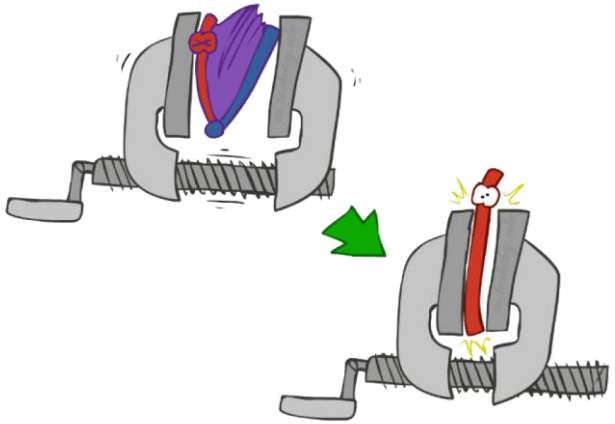
$$P(\boldsymbol{Q} \mid \boldsymbol{e})$$

- Step 1: Select the entries consistent with the evidence



- Step 2: Sum out $\boldsymbol{H}$ from model to get joint of query and evidence

$$P(\boldsymbol{Q}, \boldsymbol{e}) = \Sigma_{\boldsymbol{h}} \underbrace{P(\boldsymbol{Q}, \boldsymbol{h}, \boldsymbol{e})}_{X_1, \ldots, X_n}$$



- Step 3: Normalize

$$P(\boldsymbol{Q} \mid \boldsymbol{e}) = \alpha\, P(\boldsymbol{Q}, \boldsymbol{e})$$

Inference by Enumeration

- P(W)?

Season	Temp	Weather	P
summer	hot	sun	0.35
summer	hot	rain	0.01
summer	hot	fog	0.01
summer	hot	meteor	0.00
summer	cold	sun	0.10
summer	cold	rain	0.05
summer	cold	fog	0.09
summer	cold	meteor	0.00
winter	hot	sun	0.10
winter	hot	rain	0.01
winter	hot	fog	0.02
winter	hot	meteor	0.00
winter	cold	sun	0.15
winter	cold	rain	0.20
winter	cold	fog	0.18
winter	cold	meteor	0.00

Inference by Enumeration

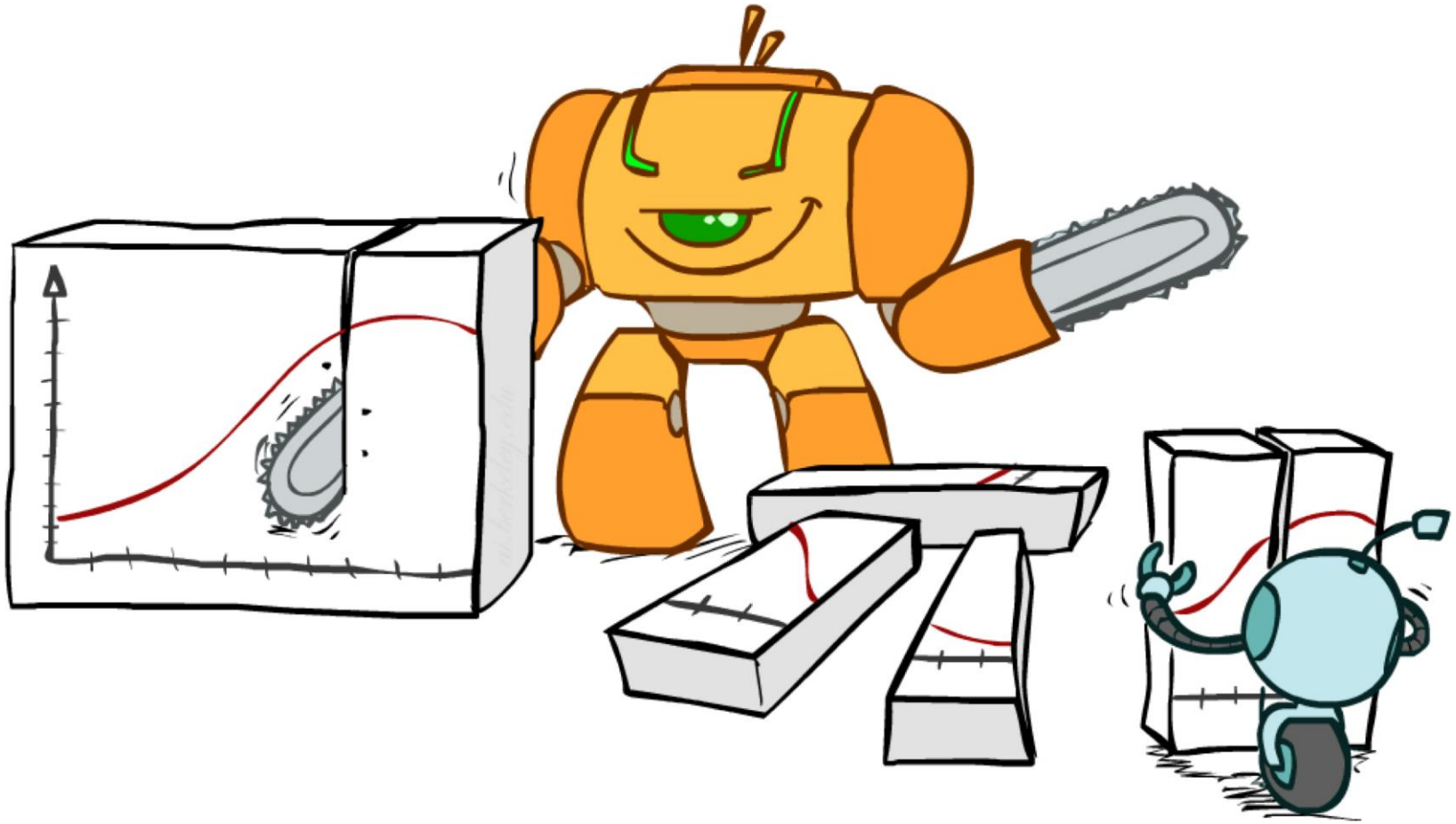
- P(W)?
- P(W | winter)?

Season	Temp	Weather	P
summer	hot	sun	0.35
summer	hot	rain	0.01
summer	hot	fog	0.01
summer	hot	meteor	0.00
summer	cold	sun	0.10
summer	cold	rain	0.05
summer	cold	fog	0.09
summer	cold	meteor	0.00
winter	hot	sun	0.10
winter	hot	rain	0.01
winter	hot	fog	0.02
winter	hot	meteor	0.00
winter	cold	sun	0.15
winter	cold	rain	0.20
winter	cold	fog	0.18
winter	cold	meteor	0.00

Inference by Enumeration

- Obvious problems:
  - Worst-case time complexity $O(d^n)$ (exponential in #hidden variables)
  - Space complexity $O(d^n)$ to store the joint distribution
  - $O(d^n)$ data points to estimate the entries in the joint distribution

Bayes Rule



Bayes' Rule

- Write the product rule both ways:

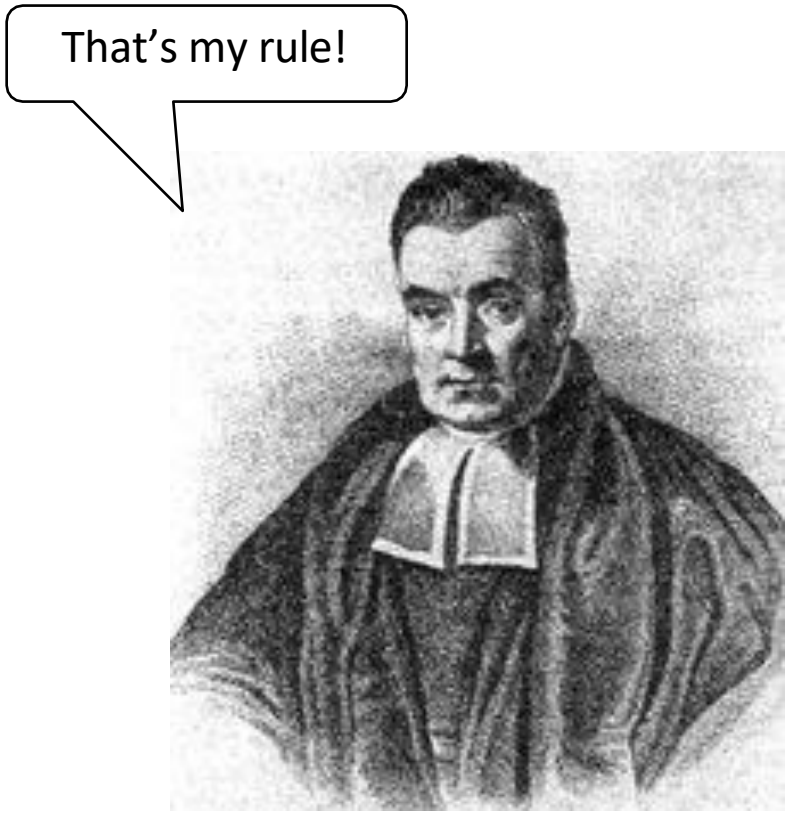
$$P(a \mid b)\, P(b) = P(a, b) = P(b \mid a)\, P(a)$$

- Dividing left and right expressions, we get:

$$P(a \mid b) = \frac{P(b \mid a)\, P(a)}{P(b)}$$

- Why is this at all helpful?
  - Lets us build one conditional from its reverse
  - Often one conditional is tricky but the other one is simple
  - Describes an "update" step from prior $P(a)$ to posterior $P(a \mid b)$
  - Foundation of many systems we'll see later (e.g. ASR, MT)

- In the running for most important AI equation!



Inference with Bayes' Rule

- Example: Diagnostic probability from causal probability:

$$P(cause \mid effect) = \frac{P(effect \mid cause)\, P(cause)}{P(effect)}$$

- Example:
  - M: meningitis, S: stiff neck

$$\left.\begin{array}{l} P(s \mid m) = 0.8 \\ P(m) = 0.0001 \\ P(s) = 0.01 \end{array}\right\} \text{Example givens}$$

$$P(m \mid s) = \frac{P(s \mid m)\, P(m)}{P(s)} = \frac{0.8 \times 0.0001}{0.01}$$

  - Note: posterior probability of meningitis still very small: 0.008 (80x bigger – why?)
  - Note: you should still get stiff necks checked out! Why?

Independence

- Two variables X and Y are (absolutely) ***independent*** if

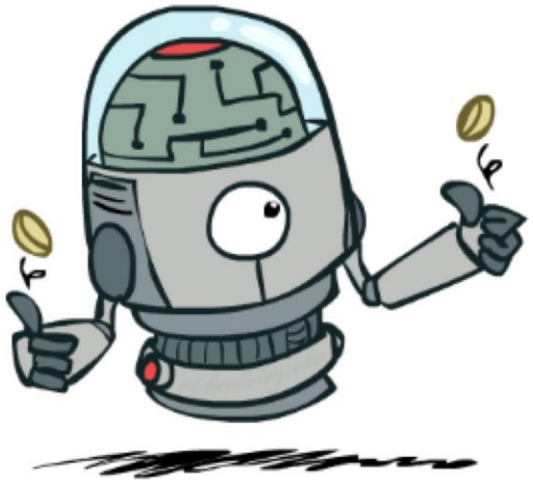
$$\forall x, y \qquad P(x, y) = P(x)\, P(y)$$

  - I.e., the joint distribution ***factors*** into a product of two simpler distributions

- Equivalently, via the product rule $P(x,y) = P(x|y)P(y)$,

$$P(x \mid y) = P(x) \quad \text{or} \quad P(y \mid x) = P(y)$$

- Example: two dice rolls $Roll_1$ and $Roll_2$
  - $P(Roll_1=5, Roll_2=3) = P(Roll_1=5)\, P(Roll_2=3) = 1/6 \times 1/6 = 1/36$
  - $P(Roll_2=3 \mid Roll_1=5) = P(Roll_2=3)$



Example: Independence

- n fair, **independent** coin flips:

$P(X_1)$

H	0.5
T	0.5

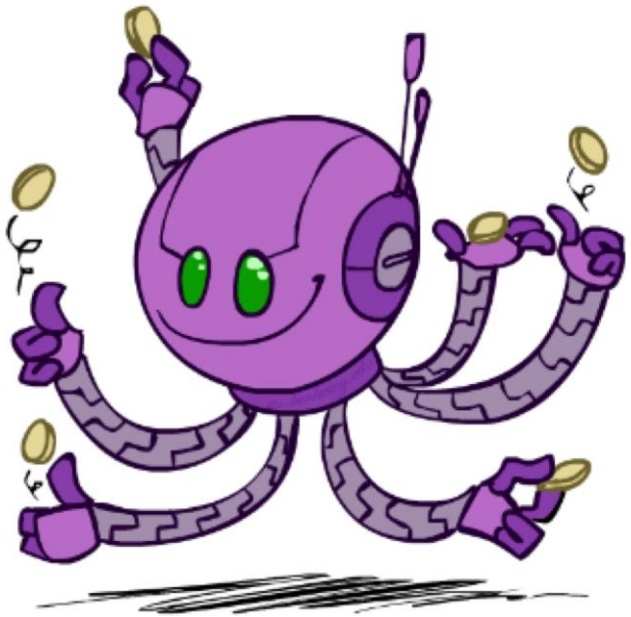
$P(X_2)$

H	0.5
T	0.5

. . .

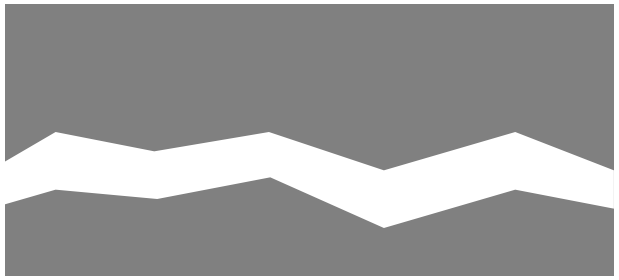
$P(X_n)$

H	0.5
T	0.5

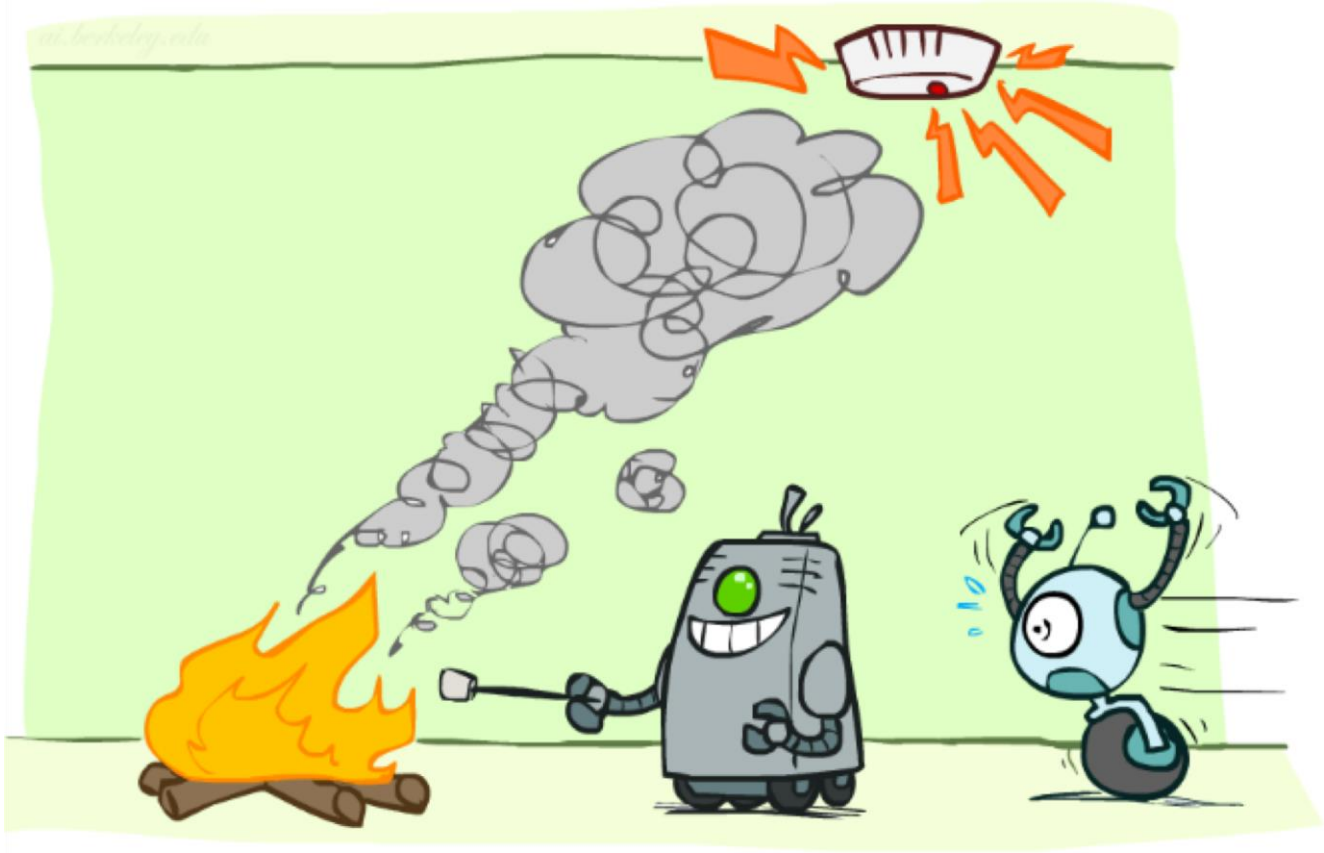


$P(X_1, X_2, \ldots, X_n)$

2^n

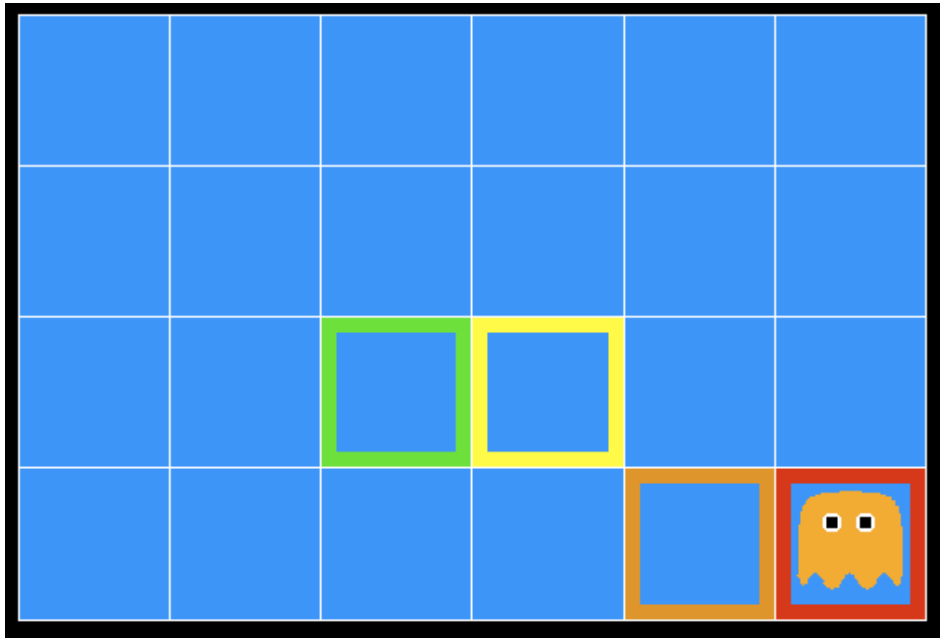


Conditional Independence



Ghostbusters

- A ghost is in the grid somewhere
- Sensor readings tell how close a square is to the ghost
  - On the ghost: usually red
  - 1 or 2 away: mostly orange
  - 3 or 4 away: typically yellow
  - 5+ away: often green
- Click on squares until confident of location, then "***bust***"



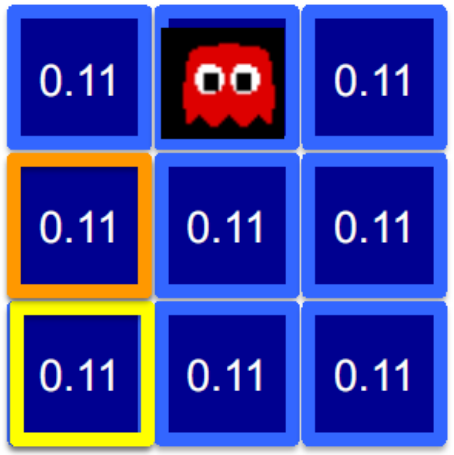
Ghostbusters model

- Variables and ranges:
  - G (ghost location) in $\{(1,1),\ldots,(3,3)\}$
  - $C_{x,y}$ (color measured at square x,y) in {red,orange,yellow,green}

0.11	0.11	0.11
0.11	0.11	0.11
0.11	0.11	0.11

- Ghostbuster physics:
  - ***Uniform prior distribution*** over ghost location: $P(G)$
  - ***Sensor model***: $P(C_{x,y} \mid G)$ (depends only on distance to G)
    - E.g. $P(C_{1,1} = \text{yellow} \mid G = (1,1)) = 0.1$

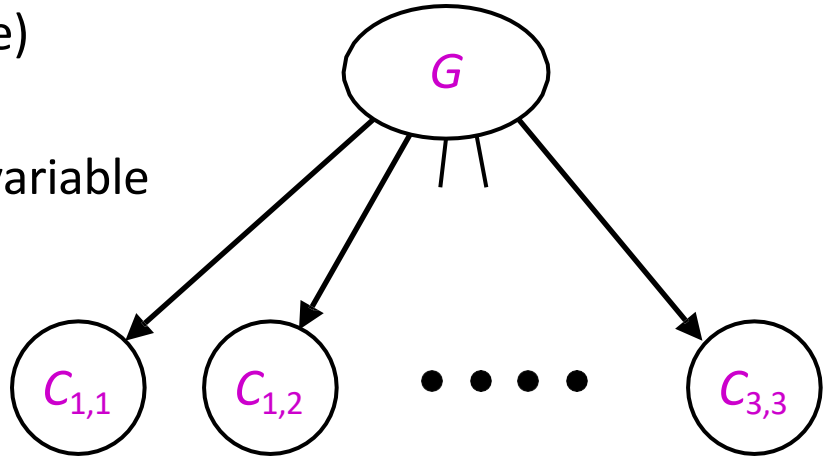
Ghostbusters model, contd.



- $P(G, C_{1,1}, \ldots C_{3,3})$ has $9 \times 4^9 = 2{,}359{,}296$ entries!!!
- Ghostbuster independence:
  - Are $C_{1,1}$ and $C_{1,2}$ **independent**?
    - E.g., does $P(C_{1,1} = \text{yellow}) = P(C_{1,1} = \text{yellow} \mid C_{1,2} = \text{orange})$?
- Ghostbuster physics again:
  - $P(C_{x,y} \mid G)$ ***depends <u>only</u> on distance to G***
    - So $P(C_{1,1} = \text{yellow} \mid \underline{G = (2,3)}) = P(C_{1,1} = \text{yellow} \mid \underline{G = (2,3)}, C_{1,2} = \text{orange})$
    - I.e., $C_{1,1}$ is ***conditionally independent*** of $C_{1,2}$ ***given*** G

Ghostbusters model, contd.

- Apply the **chain rule** to decompose the joint probability model:
- $P(G, C_{1,1}, \ldots C_{3,3}) = P(G)\, P(C_{1,1} \mid G)\, P(C_{1,2} \mid G, C_{1,1})\, P(C_{1,3} \mid G, C_{1,1}, C_{1,2}) \ldots P(C_{3,3} \mid G, C_{1,1}, \ldots, C_{3,2})$
- Now simplify using **conditional independence**:
- $P(G, C_{1,1}, \ldots C_{3,3}) = P(G)\, P(C_{1,1} \mid G)\, P(C_{1,2} \mid G)\, P(C_{1,3} \mid G) \ldots P(C_{3,3} \mid G)$
- I.e., conditional independence properties of ghostbuster physics simplify the probability model from ***exponential*** to ***quadratic*** in the number of squares
- This is called a ***Naïve Bayes*** model (We encountered this before, right?):
  - One discrete query variable (often called the ***class*** or ***category*** variable)
  - All other variables are (potentially) evidence variables
  - Evidence variables are all **conditionally** independent **given** the query variable



Conditional Independence

- ***Conditional independence*** is our most basic and robust form of knowledge about uncertain environments.

- *X* is conditionally independent of *Y* given *Z* if and only if:

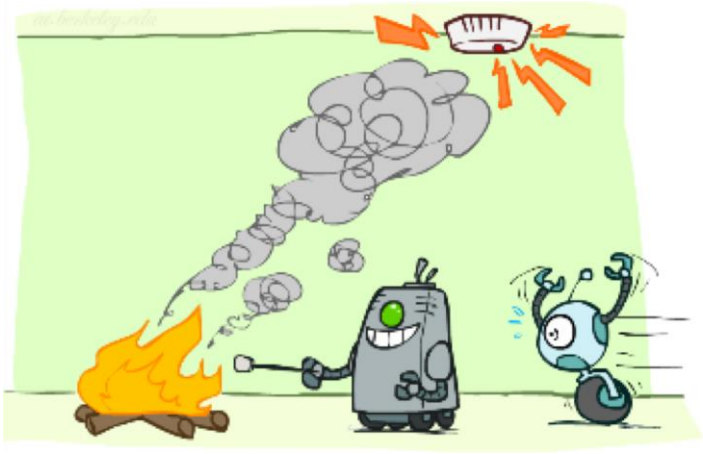
$$\forall x, y, z \;\; P(x \mid y, z) \;=\; P(x \mid z)$$

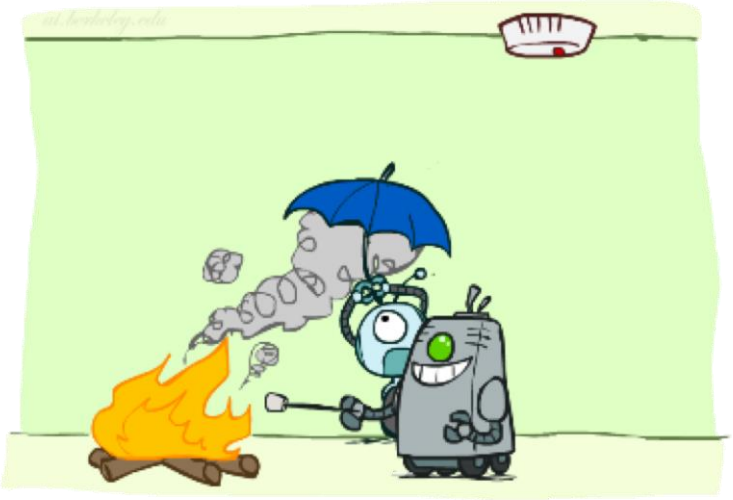
or, equivalently, if and only if

$$\forall x, y, z \;\; P(x, y \mid z) \;=\; P(x \mid z)\, P(y \mid z)$$

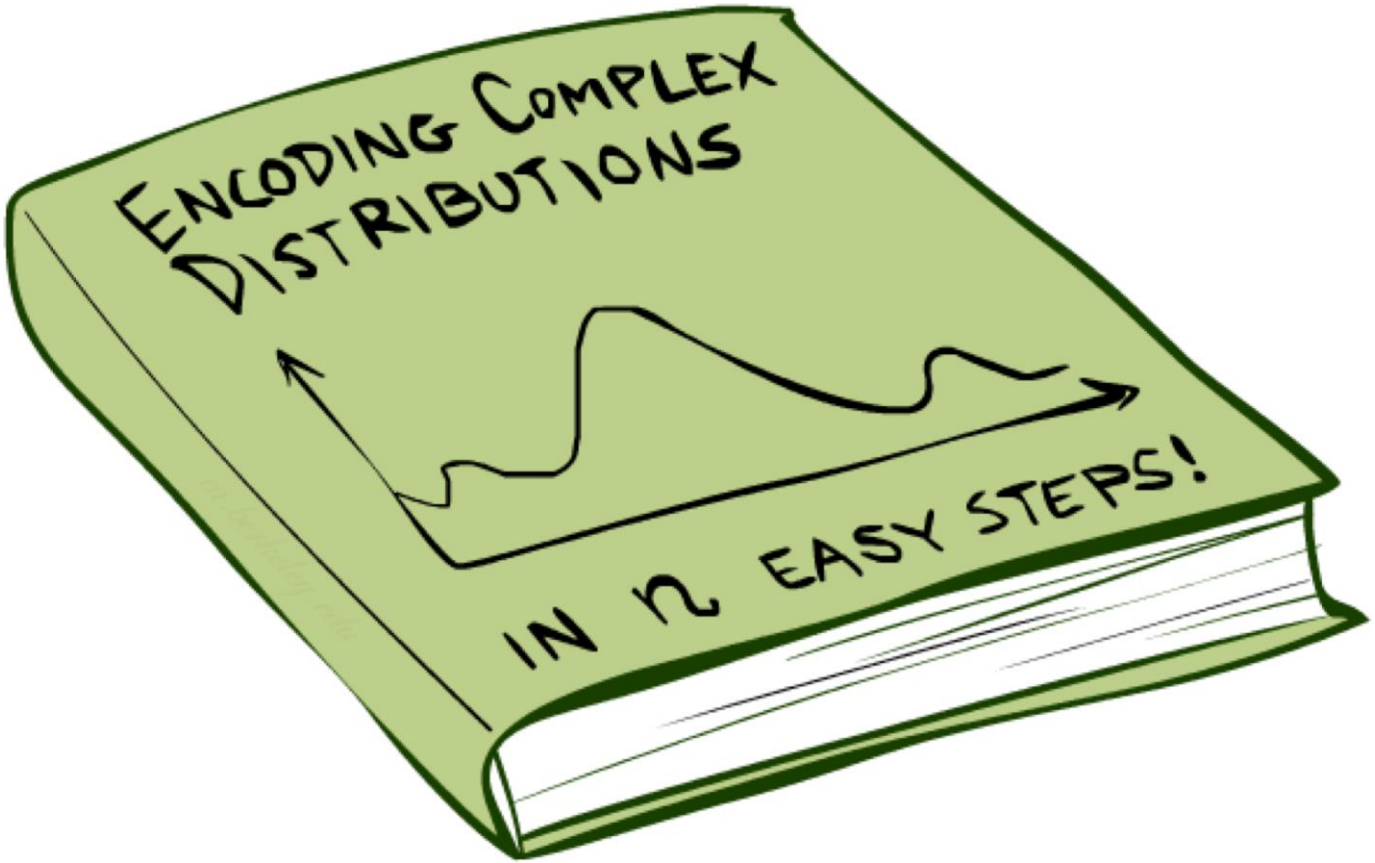
Conditional Independence

- What about this domain:
  - Fire
  - Smoke
  - Alarm





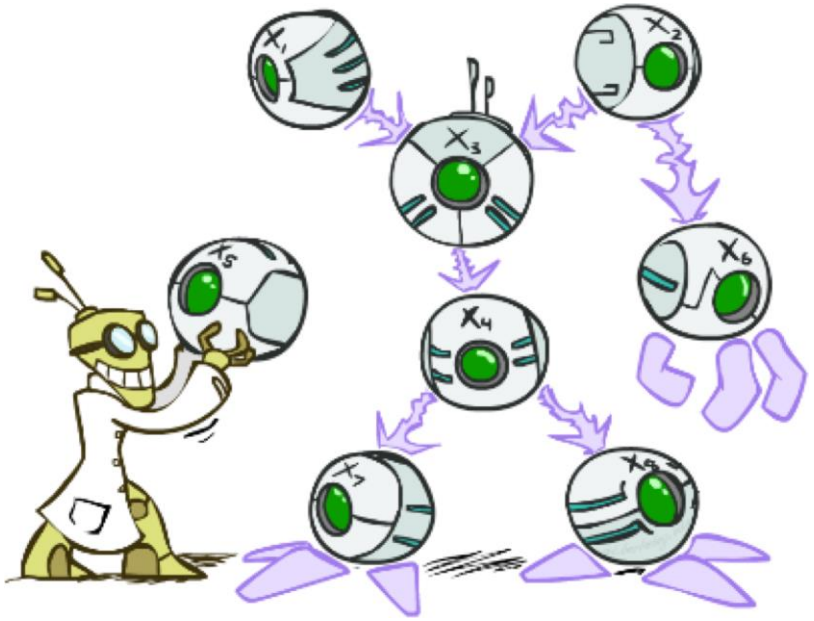
Bayes Nets: Big Picture



Bayes Nets: Big Picture

- Bayes nets: a technique for describing complex joint distributions (models) using simple, conditional distributions
  - A subset of the general class of graphical models

- Use local causality/conditional independence:
  - the world is composed of many variables,
  - each interacting **locally** with a few others



Bayes Nets

Part I: Representation

Part II: Exact inference

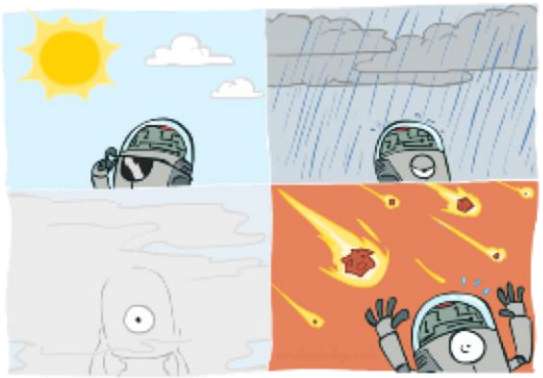
- Enumeration (always exponential complexity)
- Variable elimination (worst-case exponential complexity, often better)
- Inference is NP-hard in general

Part III: Approximate Inference

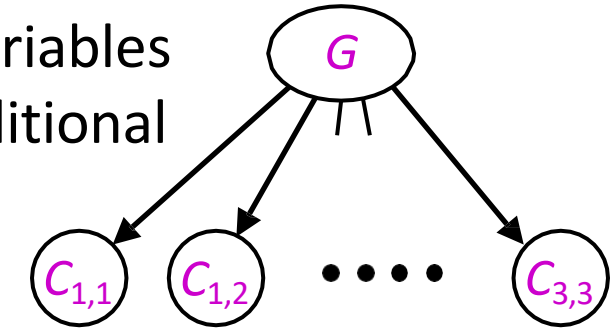
Later: Learning Bayes nets from data

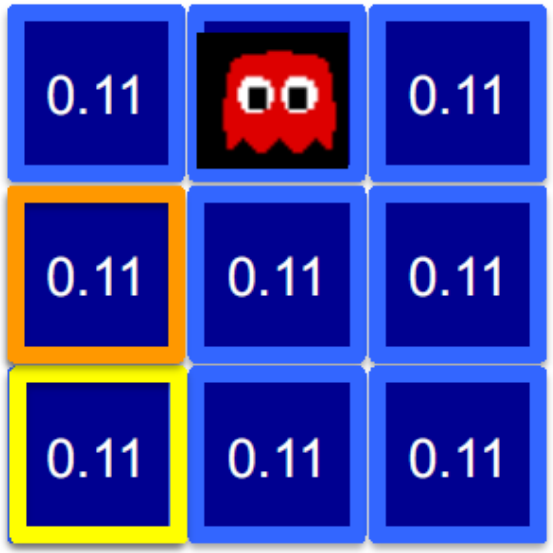
Graphical Model Notation

- Nodes: variables (with domains)
  - Can be assigned (**observed**) or unassigned (**unobserved**)



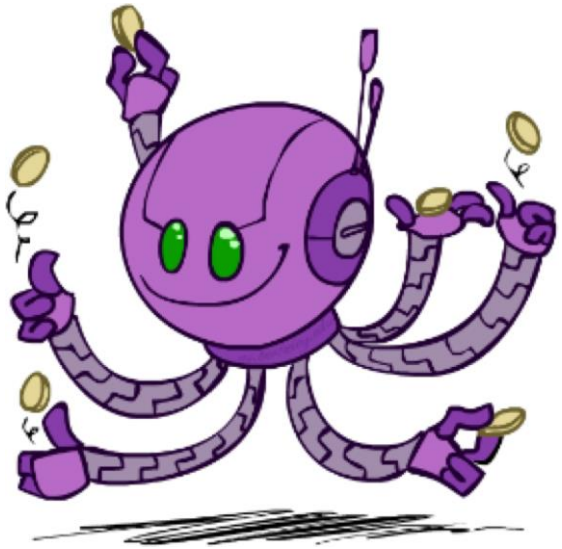
- Arcs: interactions
  - Indicate "direct influence" between variables
  - Formally: absence of arc encodes conditional independence (more later)





Example: Coin Flips

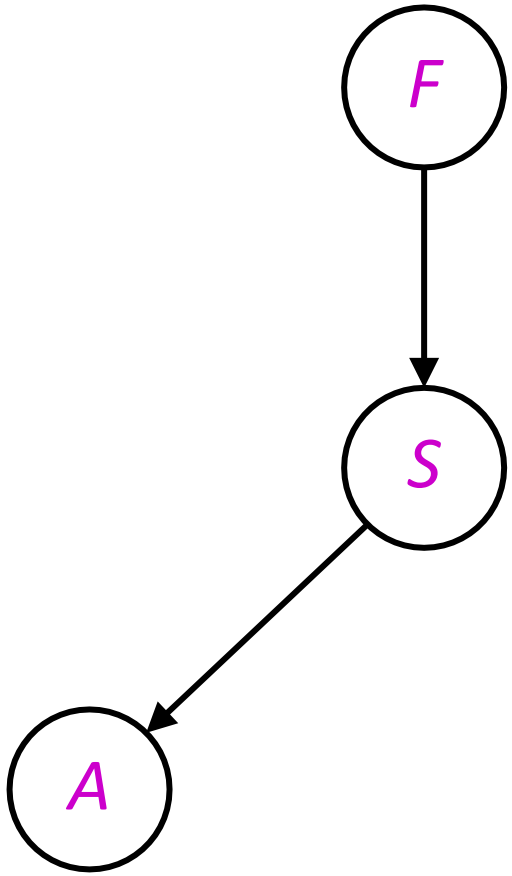
- N independent coin flips

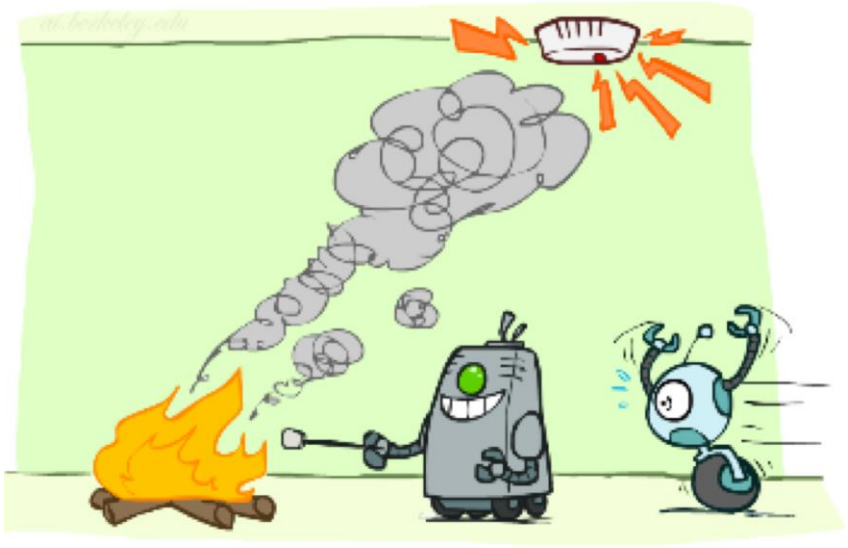


- No interactions between variables: absolute independence

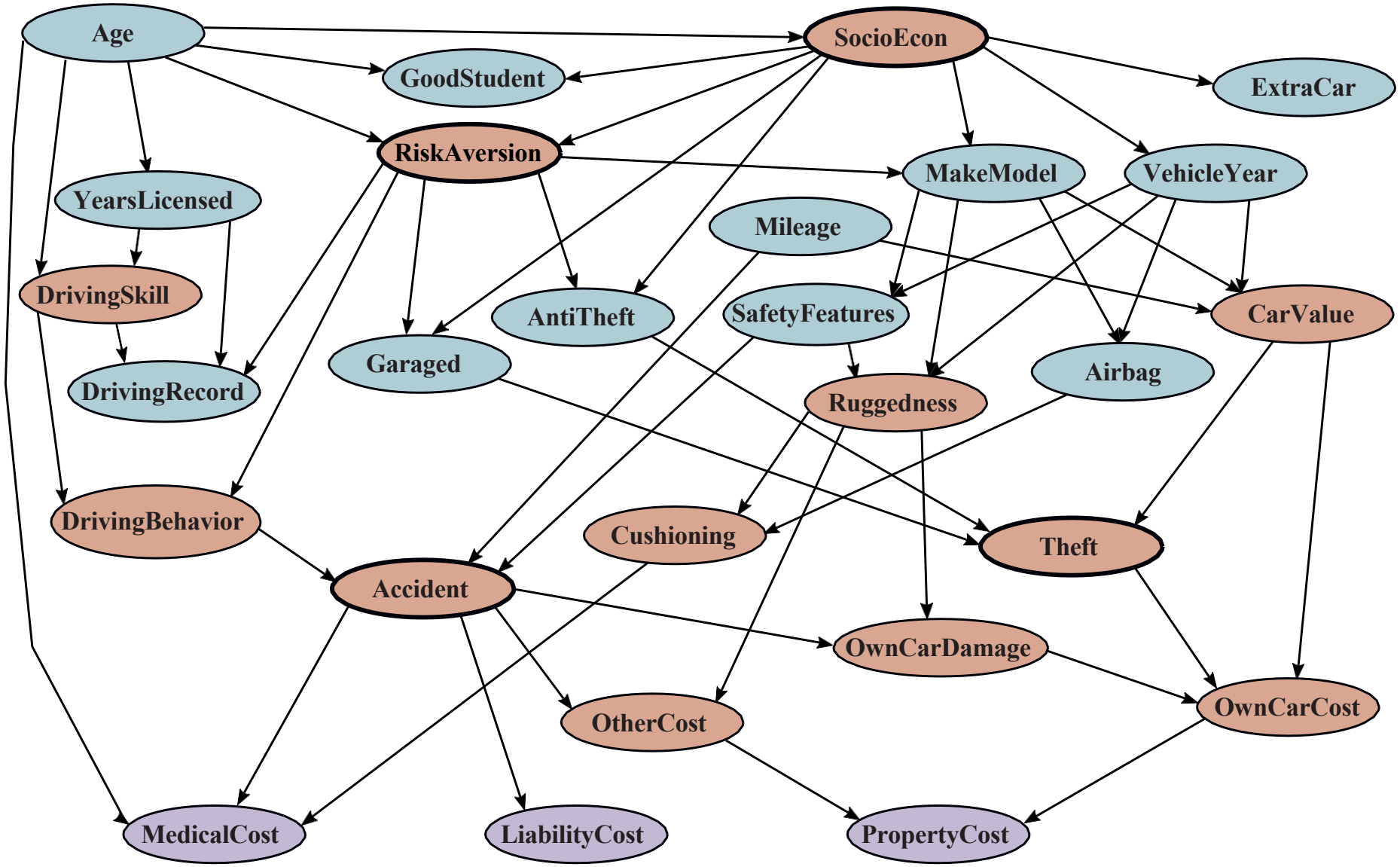
Example: Smoke alarm

- Variables:
  - F: There is fire
  - S: There is smoke
  - A: Alarm sounds

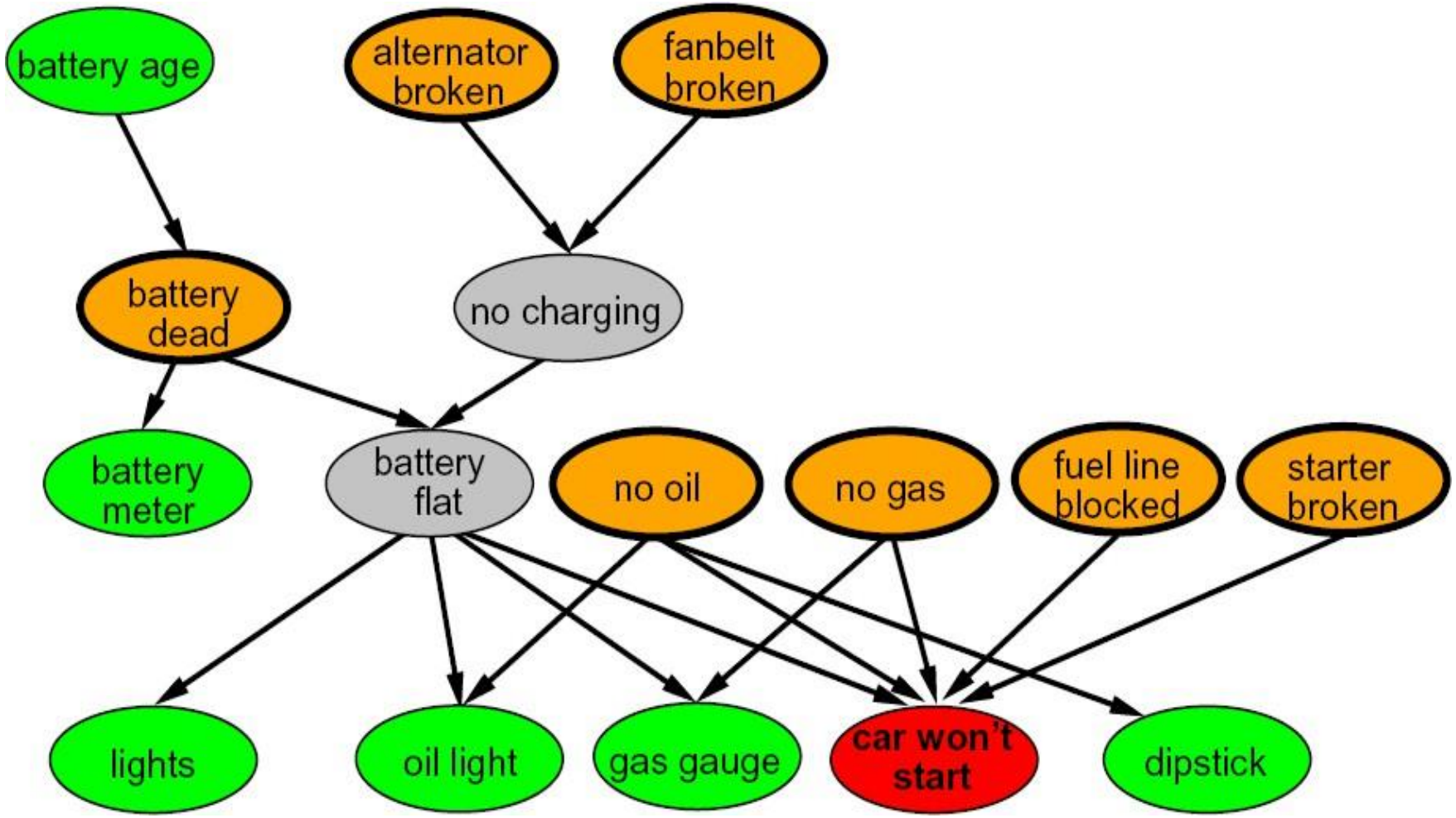




Example Bayes' Net: Car Insurance



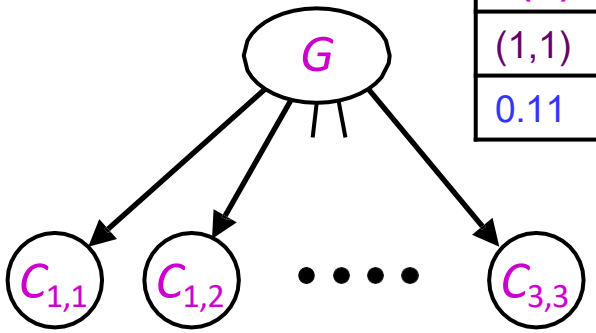
Example Bayes' Net: Car Won't Start



Bayes Net Syntax and Semantics

Bayes Net Syntax

- A set of nodes, one per variable X_i
- A **directed**, **acyclic** graph
- A **conditional** distribution for each node given its ***parent variables*** in the graph
  - ***CPT*** (conditional probability table); each row is a distribution for child given values of its parents



P(*G*)			
(1,1)	(1,2)	(1,3)	…
0.11	0.11	0.11	…

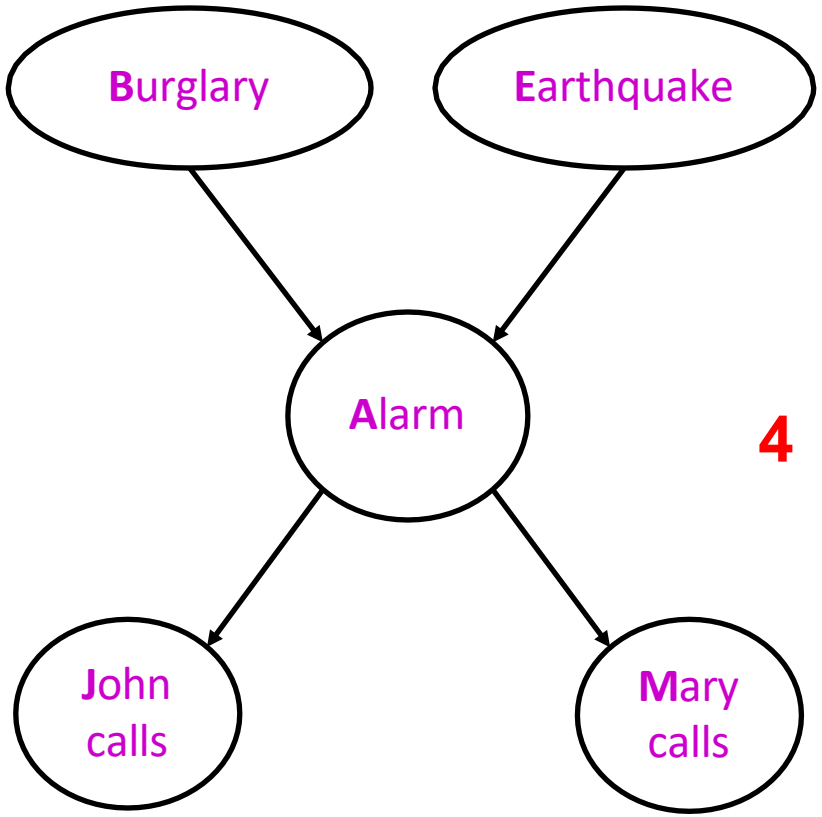
G	P($C_{1,1}$ | *G*)			
	g	y	o	r
(1,1)	0.01	0.1	0.3	0.59
(1,2)	0.1	0.3	0.5	0.1
(1,3)	0.3	0.5	0.19	0.01
…				

Bayes net = Topology (graph) + Local Conditional Probabilities

Example: Alarm Network

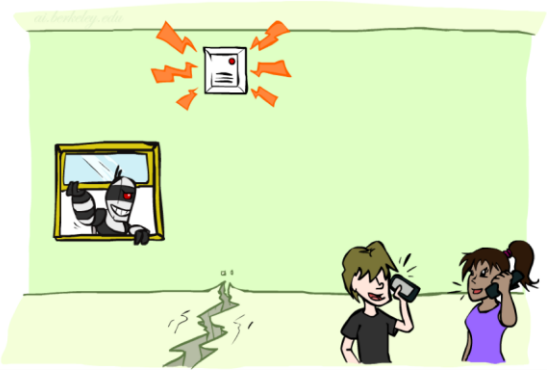
P(B)	
true	false
0.001	0.999

1



1

P(E)	
true	false
0.002	0.998



B	E	P(A|B,E)	
		true	false
true	true	0.95	0.05
true	false	0.94	0.06
false	true	0.29	0.71
false	false	0.001	0.999

4

A	P(J|A)	
	true	false
true	0.9	0.1
false	0.05	0.95

2

A	P(M|A)	
	true	false
true	0.7	0.3
false	0.01	0.99

2

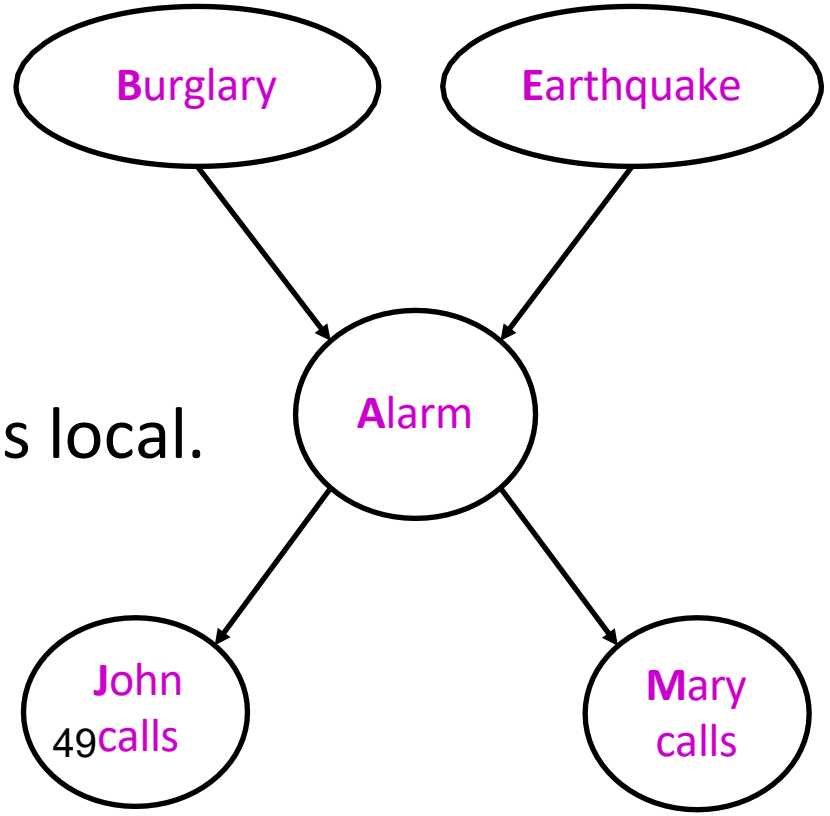
Number of ***free parameters*** in each CPT:

Parent range sizes $d_1,\ldots,d_k$

Child range size d
Each table row must sum to 1

General formula for sparse BNs

- Suppose
  - n variables
  - Maximum range size is d
  - Maximum number of parents is k
- Full joint distribution has size $O(d^n)$
- Bayes net has size $O(n \cdot d^k)$
  - Linear scaling with n as long as causal structure is local.



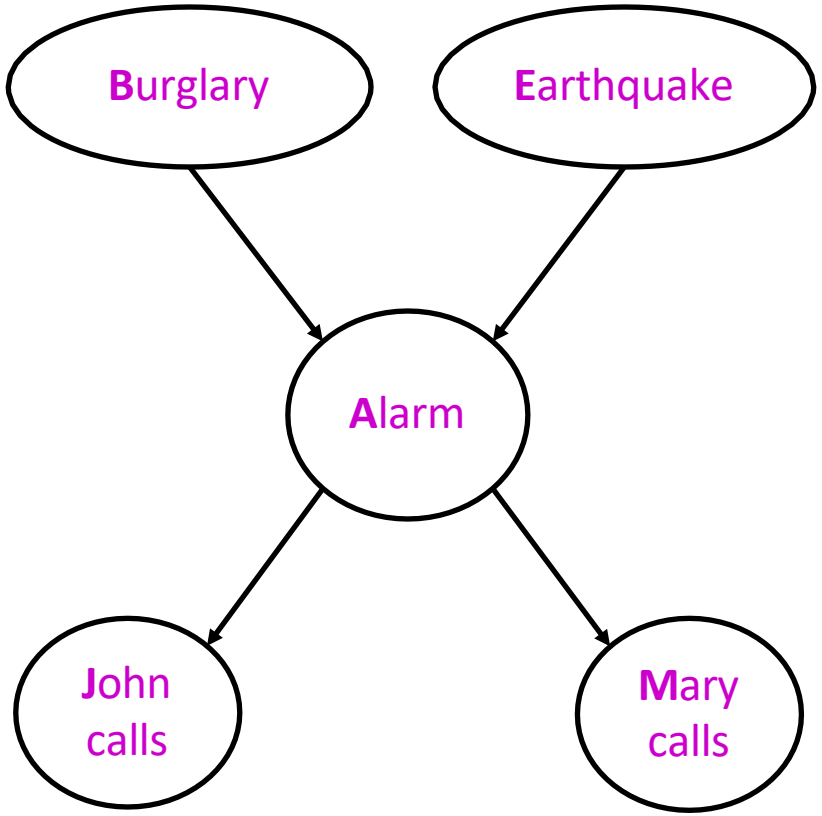
Bayes net global semantics

- Bayes nets encode **joint** distributions as product of **conditional** distributions on each variable:

$$P(X_1,..,X_n) \;=\; \Pi_i \; P(X_i \mid Parents(X_i))$$

Example

P(B)	
true	false
0.001	0.999



P(E)	
true	false
0.002	0.998

$P(b, \neg e, a, \neg j, \neg m) =$

$P(b)\ P(\neg e)\ P(a|b,\neg e)\ P(\neg j|a)\ P(\neg m|a)$

$=.001x.998x.94x.1x.3=.000028$

B	E	P(A|B,E)	
		true	false
true	true	0.95	0.05
true	false	0.94	0.06
false	true	0.29	0.71
false	false	0.001	0.999

A	P(J|A)	
	true	false
true	0.9	0.1
false	0.05	0.95

A	P(M|A)	
	true	false
true	0.7	0.3
false	0.01	0.99

Conditional independence in BNs

- Compare the Bayes net global semantics

$$P(X_1,..,X_n) = \Pi_i \, P(X_i \mid Parents(X_i))$$

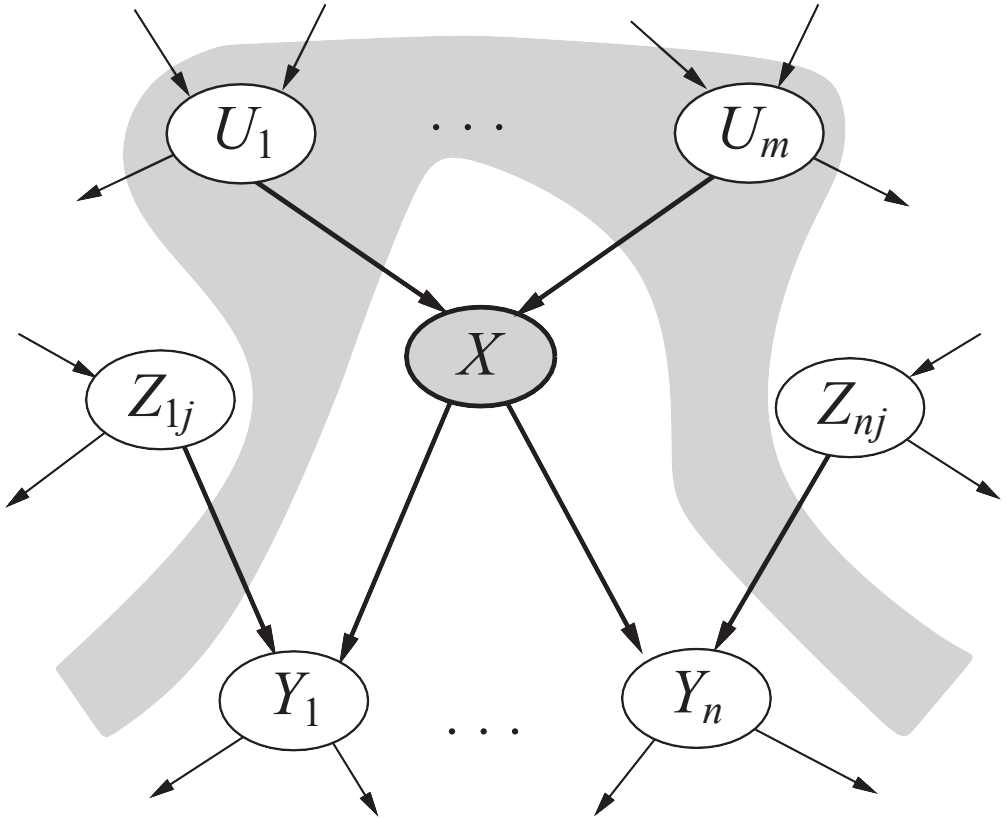
with the **chain rule** identity

$$P(X_1,..,X_n) = \Pi_i \, P(X_i \mid X_1,\ldots,X_{i-1})$$

- Assume (without loss of generality) that $X_1,..,X_n$ sorted in topological order according to the graph (i.e., parents before children), so $Parents(X_i) \subseteq X_1, \ldots, Xi_1$
- So the Bayes net asserts conditional independences $P(X_i \mid X_1,\ldots,X_{i-1}) = P(X_i \mid Parents(X_i))$
  - To ensure these are valid, choose parents for node X_i that "shield" it from other predecessors

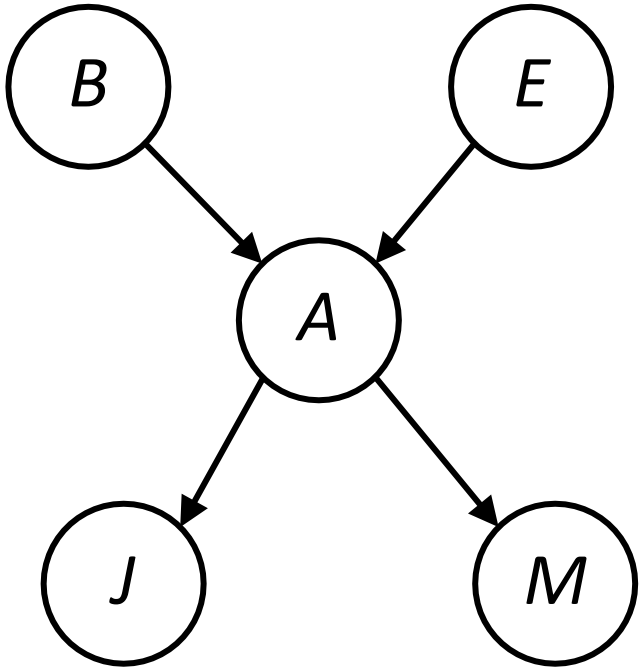
Conditional independence semantics

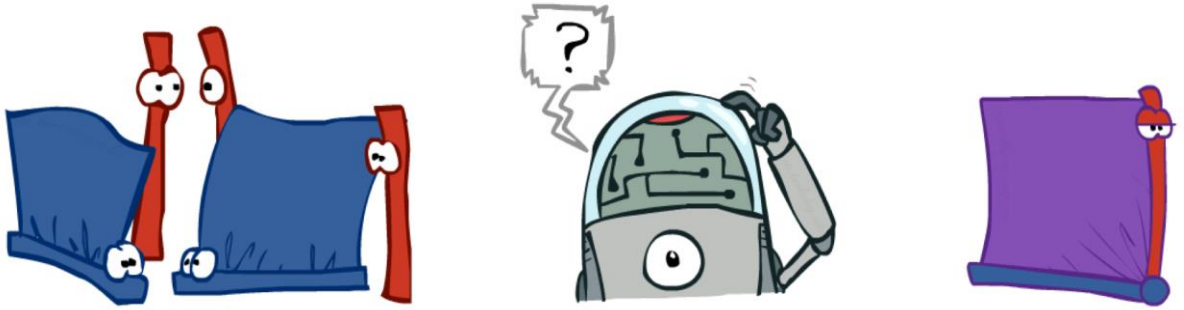
- ***Every variable is conditionally independent of its non-descendants given its parents***
- Conditional independence semantics <=> global semantics



Inference by Enumeration in Bayes Net

- Reminder of inference by enumeration:
  - Any probability of interest can be computed by summing entries from the joint distribution: $P(\boldsymbol{Q} \mid \boldsymbol{e}) = \boldsymbol{\alpha} \sum_{\boldsymbol{h}} P(\boldsymbol{Q}, \boldsymbol{h}, \boldsymbol{e})$
  - Entries from the joint distribution can be obtained from a BN by multiplying the corresponding conditional probabilities
- $P(B \mid j, m) = \alpha \sum_{e,a} P(B, e, a, j, m)$
- $= \alpha \sum_{e,a} P(B)\, P(e)\, P(a \mid B, e)\, P(j \mid a)\, P(m \mid a)$
- So inference in Bayes nets means computing sums of products of numbers: sounds easy!!
- Problem: sums of ***exponentially many*** products!
- We have method to do it more efficiently.





Bayes Nets

✔ Part I: Representation

Part II: Exact inference

- ✔ Enumeration (always exponential complexity)
- Variable elimination (worst-case exponential complexity, often better)
- Inference is NP-hard in general

Part III: Approximate Inference

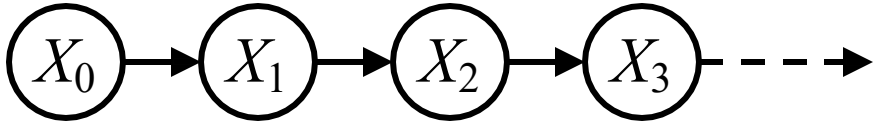
Later: Learning Bayes nets from data

Uncertainty and Time

- Often, we want to reason about a ***sequence*** of observations where the state of the underlying system is ***changing***
  - Speech recognition
  - Robot localization
  - User attention
  - Medical monitoring
  - Global climate
- Need to introduce **time** into our models

Markov Models (aka Markov chain/process)

- Value of X at a given time is called the ***state*** (usually discrete, finite)



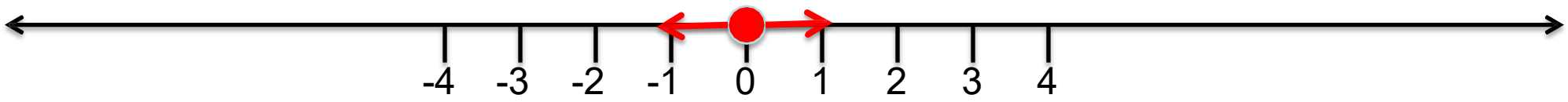
$P(X_0)$ $\qquad$ $P(X_t \mid X_{t-1})$

- The ***transition model*** $P(X_t \mid X_{t-1})$ specifies how the state evolves over time
- ***Stationarity*** assumption: transition probabilities are the same at all times
- ***Markov*** assumption: "future is **independent** of the past **given** the present"
  - X_{t+1} is independent of $X_0, \ldots, X_{t-1}$ given X_t
  - This is a ***first-order*** Markov model (a kth-order model allows dependencies on k earlier steps)
- Joint distribution $P(X_0, \ldots, X_T) = P(X_0) \prod_t P(X_t \mid X_{t-1})$

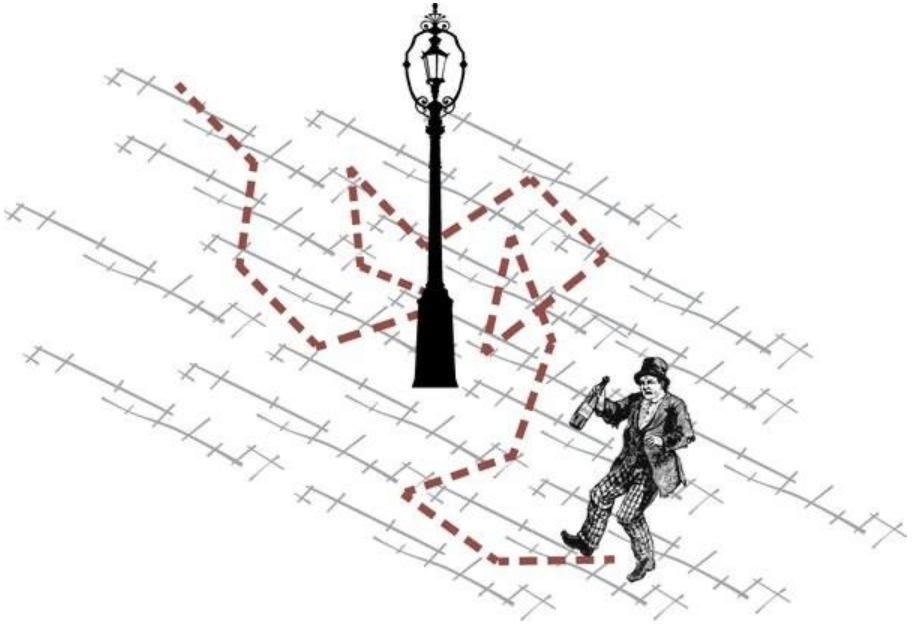
Are Markov models a special case of Bayes nets?

- Yes and no!
- Yes:
  - Directed acyclic graph, joint = product of conditionals
- No:
  - Infinitely many variables (unless we truncate)
  - Repetition of transition model not part of standard Bayes net syntax

Example: Random walk in one dimension

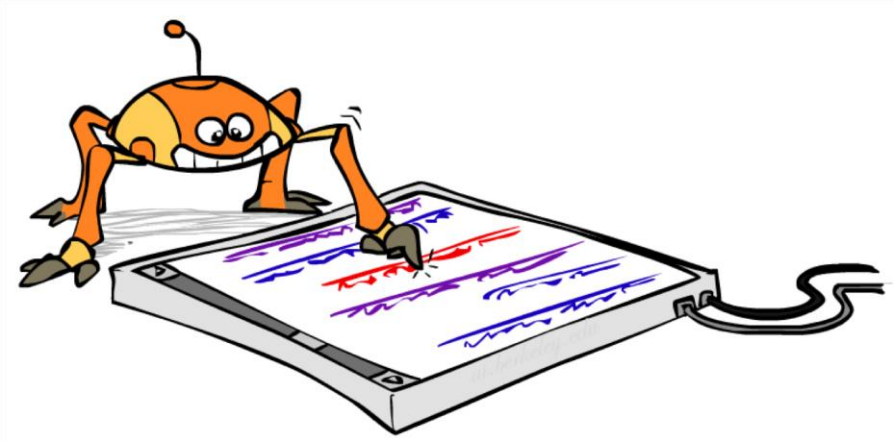


- State: location on the unbounded integer line
- Initial probability: starts at 0
- Transition model: $P(X_t = k \mid X_{t-1} = k \pm 1) = 0.5$
- Applications: particle motion in crystals,
  stock prices, gambling, genetics, etc.
- Questions:
  - How far does it get as a function of t?
    - Expected distance is $O(\sqrt{t})$



Example: Web browsing

- State: URL visited at step *t*
- Transition model:
  - With probability *p*, choose an outgoing link at random
  - With probability (1-*p*), choose an arbitrary new page
- Question: What is the ***stationary distribution*** over pages?
  - I.e., if the process runs forever, what fraction of time does it spend in any given page?
- Application: Google page rank



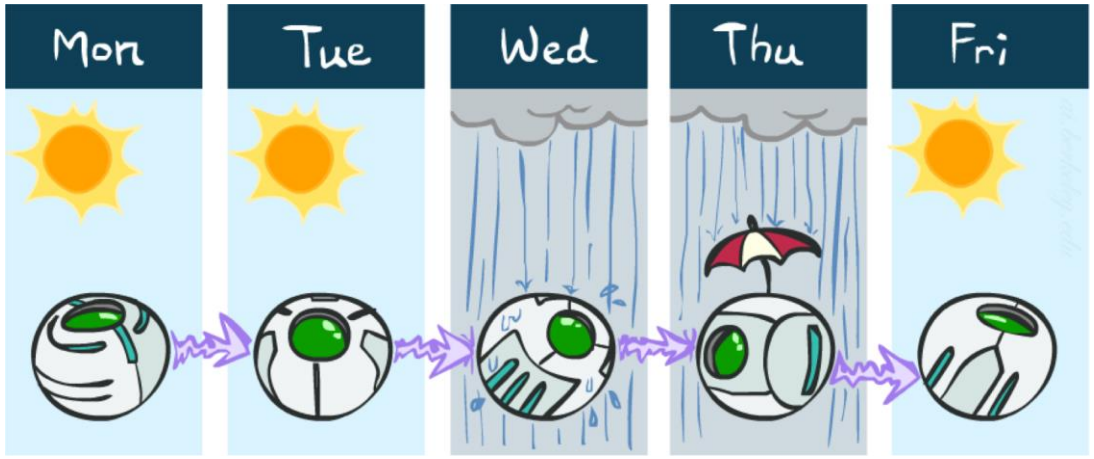
Example: Weather

- States {rain, sun}
- Initial distribution $P(X_0)$

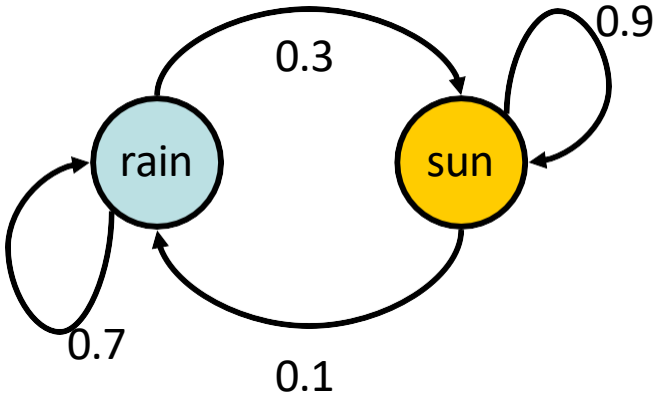
$P(X_0)$	
sun	rain
0.5	0.5

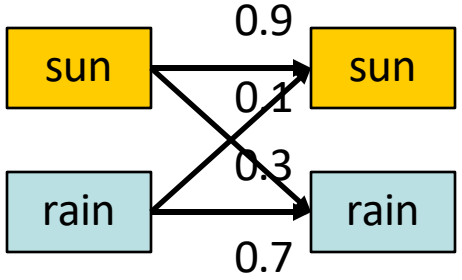
- Transition model $P(X_t \mid X_{t-1})$

X_{t-1}	$P(X_t \mid X_{t-1})$	
	sun	rain
sun	0.9	0.1
rain	0.3	0.7



Two new ways of representing the same CPT

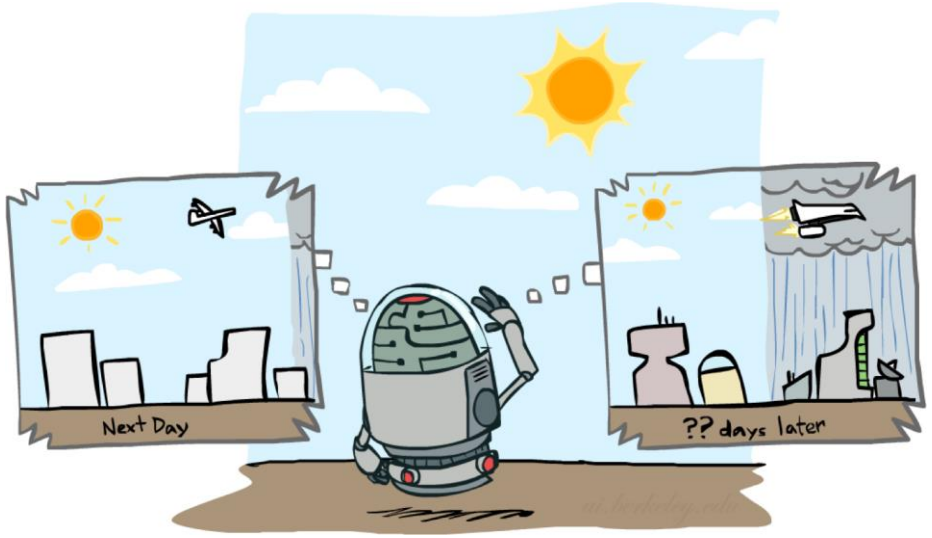




Weather prediction

- Time 0: <0.5,0.5>

X_{t-1}	$P(X_t \mid X_{t-1})$	
	sun	rain
sun	0.9	0.1
rain	0.3	0.7

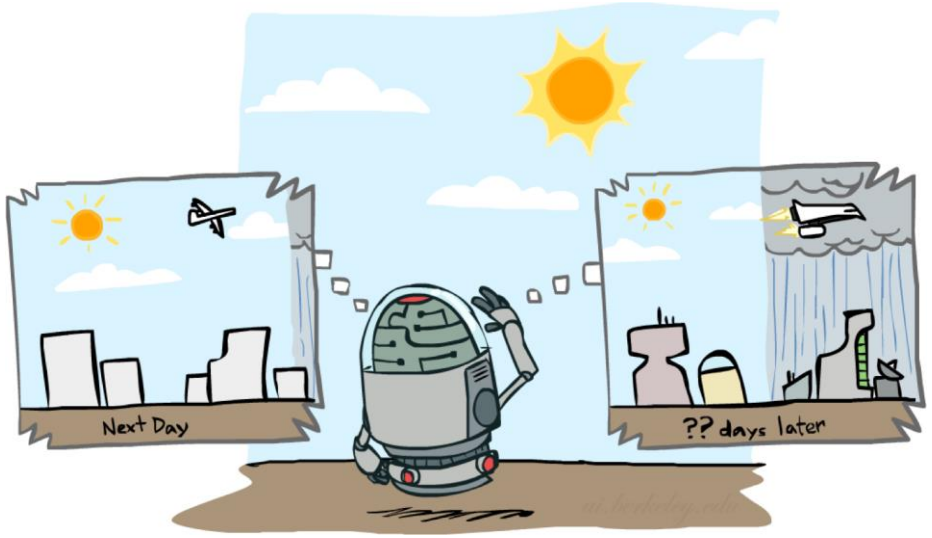


- What is the weather like at time 1?
  - $P(X_1) = \Sigma_{x_0} P(X_1, X_0 = x_0)$
  - $= \Sigma_{x_0} P(X_0 = x_0)\, P(X_1 \mid X_0 = x_0)$
  - $= 0.5\langle 0.9, 0.1\rangle + 0.5\langle 0.3, 0.7\rangle = \langle 0.6, 0.4\rangle$

Weather prediction, contd.

- Time 1: <0.6,0.4>

X_{t-1}	$P(X_t \mid X_{t-1})$	
	sun	rain
sun	0.9	0.1
rain	0.3	0.7

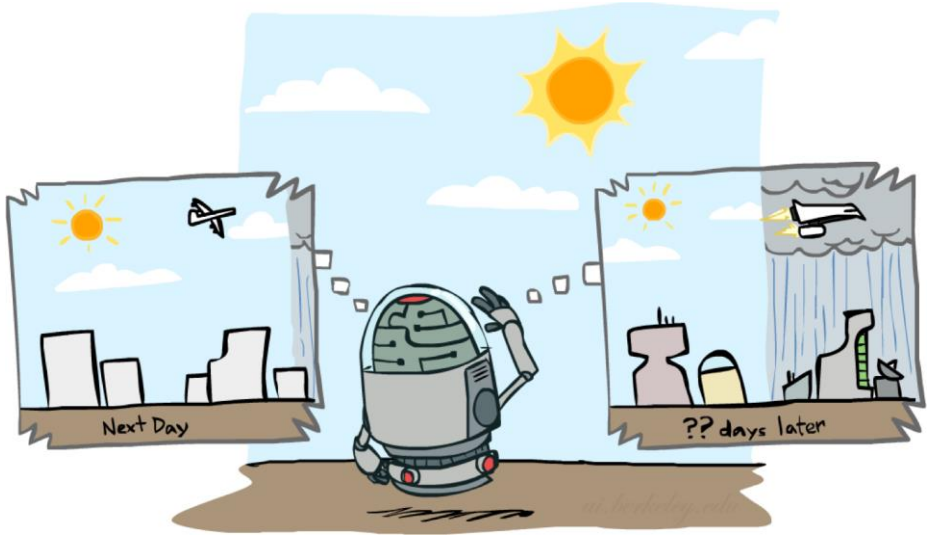


- What is the weather like at time 2?
  - $P(X_2) = \Sigma_{x_1} P(X_2, X_1 = x_1)$
  - $= \Sigma_{x_1} P(X_1 = x_1)\, P(X_2 \mid X_1 = x_1)$
  - $= 0.6\langle 0.9, 0.1\rangle + 0.4\langle 0.3, 0.7\rangle = \langle 0.66, 0.34\rangle$

Weather prediction, contd.

- Time 2: <0.66,0.34>

X_{t-1}	$P(X_t \mid X_{t-1})$	
	sun	rain
sun	0.9	0.1
rain	0.3	0.7



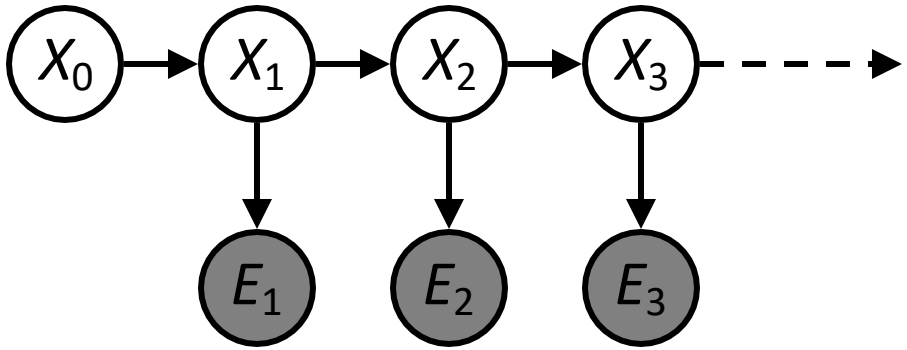
- What is the weather like at time 3?
  - $P(X_3) = \Sigma_{x_2} P(X_3, X_2 = x_2)$
  - $= \Sigma_{x_2} P(X_2 = x_2)\, P(X_3 \mid X_2 = x_2)$
  - $= 0.66\langle 0.9, 0.1\rangle + 0.34\langle 0.3, 0.7\rangle = \langle 0.696, 0.304\rangle$

Hidden Markov Models

Hidden Markov Models

- Usually the true state is not observed directly

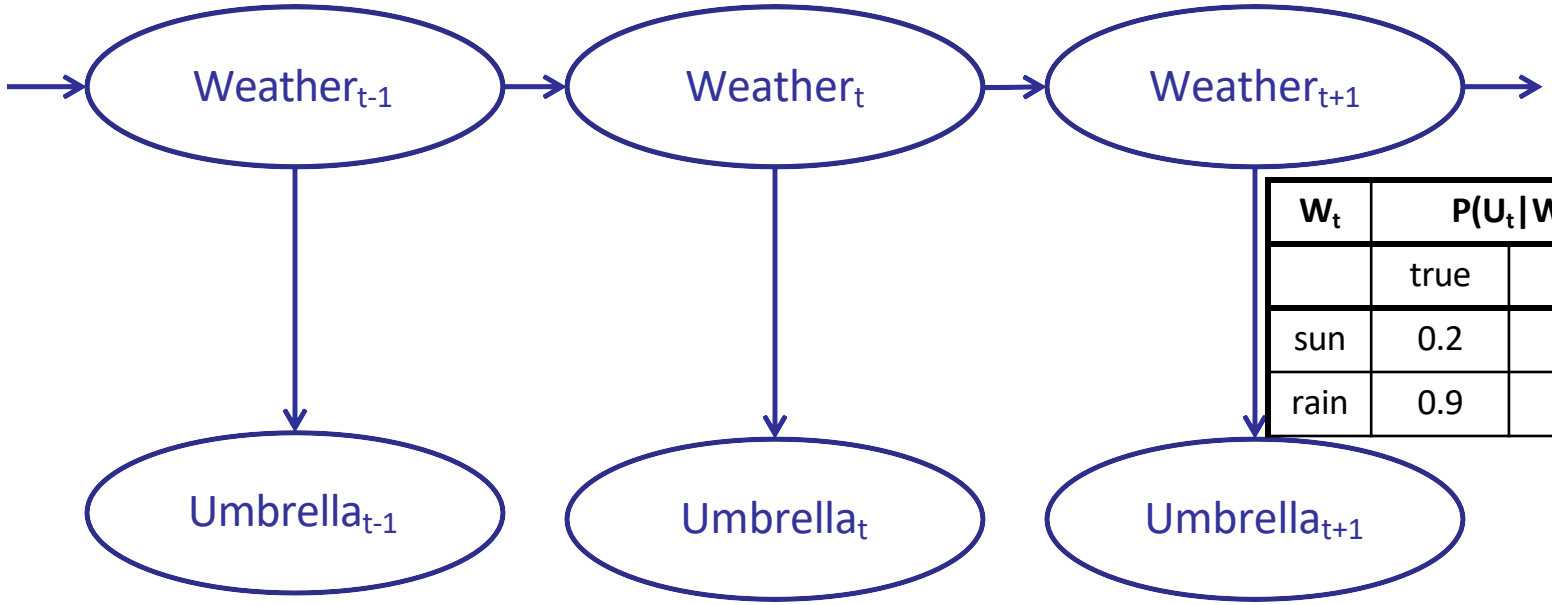
- Hidden Markov models (HMMs)
  - Underlying Markov chain over states X
  - You observe evidence E at each time step
  - X_t is a single discrete variable; E_t may be continuous and may consist of several variables



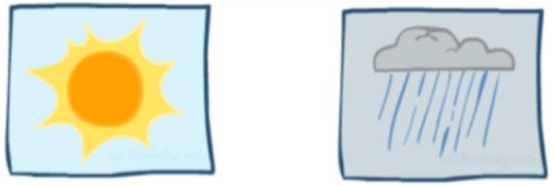
Example: Weather HMM

- An HMM is defined by:
  - Initial distribution: $P(X_0)$
  - Transition model: $P(X_t \mid X_{t-1})$
  - Sensor model: $P(E_t \mid X_t)$

W_{t-1}	$P(W_t \mid W_{t-1})$	
	sun	rain
sun	0.9	0.1
rain	0.3	0.7



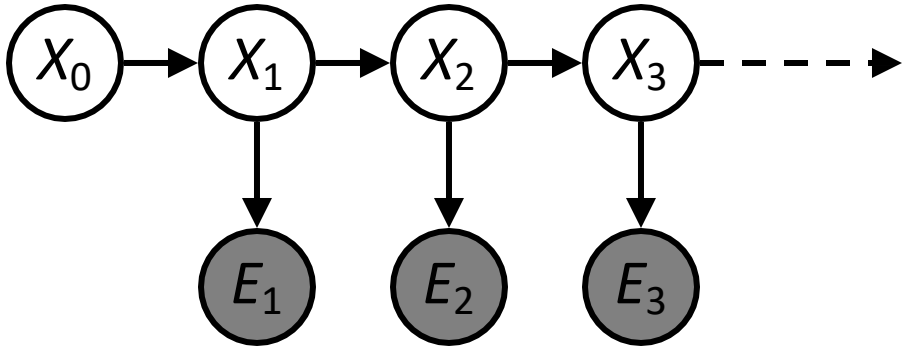
W_t	$P(U_t \mid W_t)$	
	true	false
sun	0.2	0.8
rain	0.9	0.1



HMM as probability model

- Joint distribution for **Markov** model: $P(X_0, \ldots, X_T) = P(X_0) \prod_{t=1:T} P(X_t \mid X_{t-1})$
- Joint distribution for **hidden Markov** model:

  $P(X_0, X_1, \ldots, X_T, \boldsymbol{E_T}) = P(X_0) \prod_{t=1:T} P(X_t \mid X_{t-1})\, \boldsymbol{P(E_t \mid X_t)}$
- Future states are independent of the past given the present
- Current evidence is independent of everything else given the current state
- Are evidence variables independent of each other?



Useful notation:

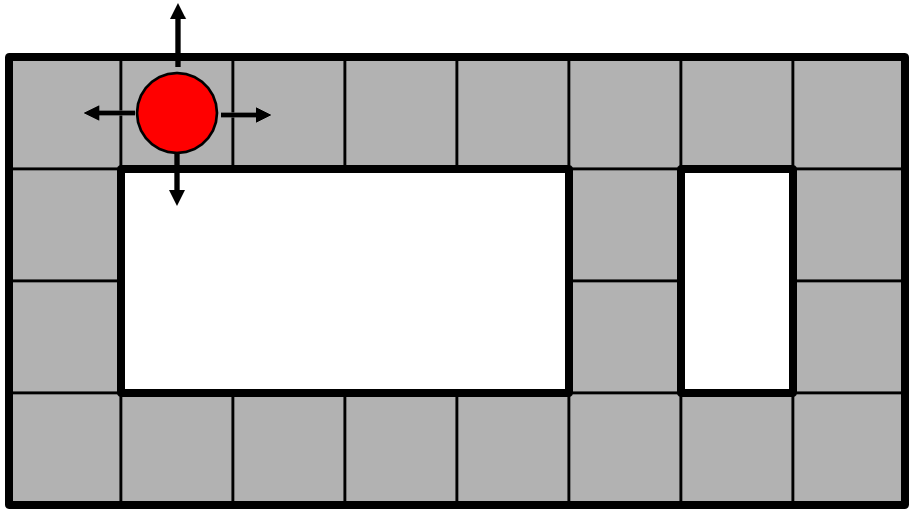
$X_{a:b} = X_a, X_{a+1}, \ldots, X_b$

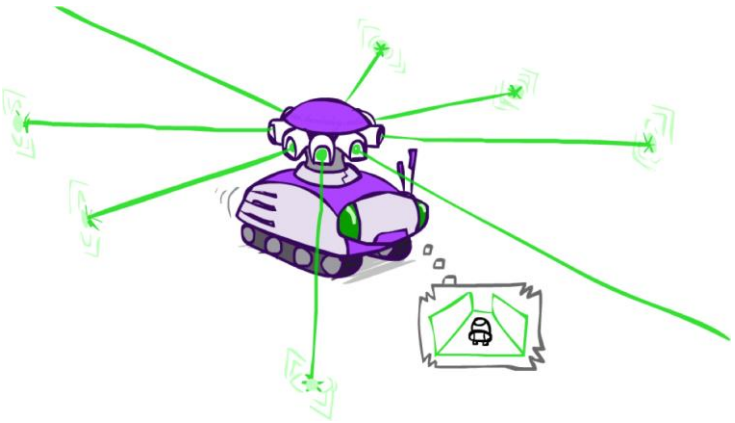
Real HMM Examples

- Speech recognition HMMs:
  - Observations are acoustic signals (continuous valued)
  - States are specific positions in specific words (so, tens of thousands)
- Machine translation HMMs:
  - Observations are words (tens of thousands)
  - States are translation options
- Robot tracking:
  - Observations are range readings (continuous)
  - States are positions on a map (continuous)
- Molecular biology:
  - Observations are nucleotides ACGT
  - States are coding/non-coding/start/stop/splice-site etc.

Example: Robot Localization

Example from
Michael Pfeiffer





Prob 0 1

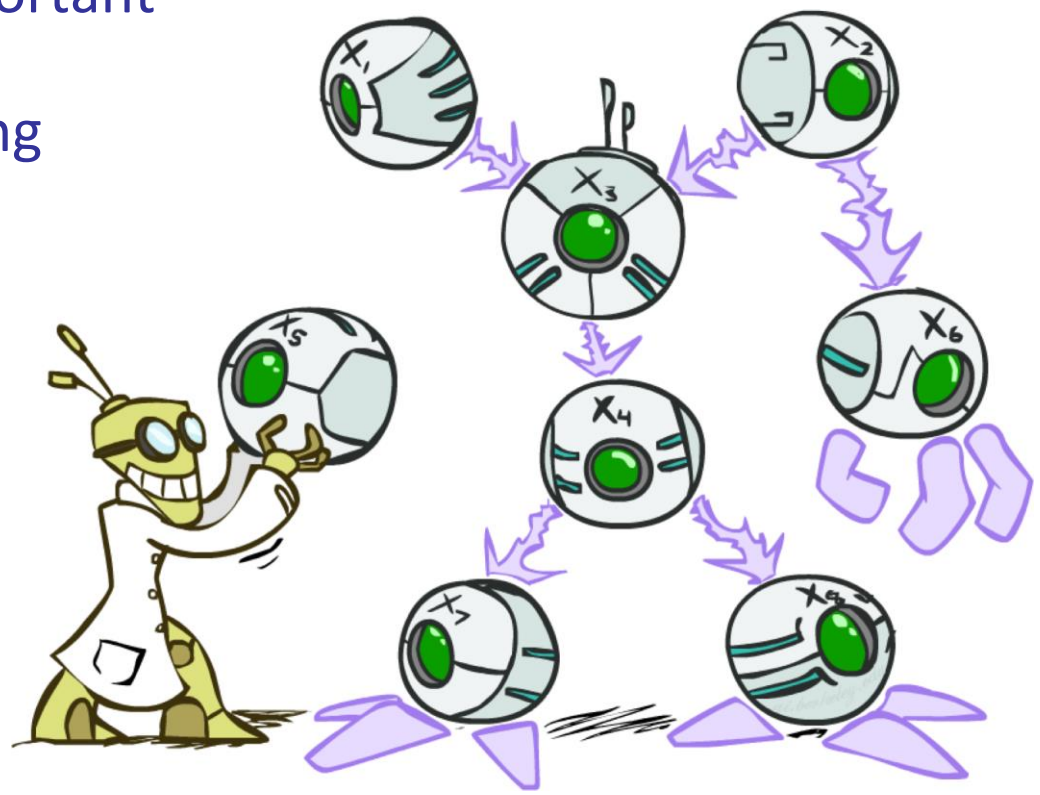
t=0

Sensor model: four bits for wall/no-wall in each direction, never more than 1 mistake

Transition model: action may fail with small prob.

Summary

- **Independence** and **conditional independence** are important forms of probabilistic knowledge
- Bayes net encode **joint** distributions efficiently by taking advantage of **conditional independence**
  - Global joint probability = product of local conditionals
- Exact inference = sums of products of conditional probabilities from the network



Acknowledgement

The lecture slides are based on the materials from ai.Berkey.edu

Questions?

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